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Machinery Dynamics
Report 2

Author

Laurenz Walter Torni - s252851

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1 Problem Statement

Rotating machines such as centrifugal compressors are subject to lateral vibrations that can lead to instability, excessive vibration amplitudes, and ultimately catastrophic failure. Understanding the dynamic behaviour of such systems is therefore essential in the design process. This report investigates the lateral dynamics and stability of a flexible rotor-bearing system modelled after a physical test rig at the Technical University of Denmark, which simulates an overhung centrifugal compressor.

The analysis is carried out using the Finite Element Method (FEM) to build mathematical models of increasing complexity. Starting from a free-free shaft model, rigid discs are progressively added, and the system is subsequently mounted on ball bearings and hydrodynamic journal bearings. At each stage, the natural frequencies and mode shapes are computed and compared against experimental modal testing data to validate and adjust the models.

The core objectives are to predict critical speeds using Campbell diagrams, assess the unbalance response at the disc location, and determine the stability threshold of the rotor-bearing system under hydrodynamic lubrication. The influence of bearing type and configuration on system stability is also examined, with the aim of identifying the most suitable bearing setup from an operational and safety perspective.

2 Mathematical Modelling

2.1 Modelling Assumptions for the Flexible Shaft

In order to model the lateral vibrations of the flexible shaft in a free–free condition, the shaft is idealised as a one–dimensional beam and discretised using Euler–Bernoulli finite elements. The real three–dimensional geometry is represented by the shaft centreline, with each element characterised by a constant circular cross–section. Local geometric details such as fillets, keyways, threads and chamfers are neglected.

Bending is described by Euler–Bernoulli theory, meaning that plane cross–sections remain plane and normal to the neutral axis during deformation, and shear deformation is not accounted for. Lateral displacements and rotations are assumed small, which allows linearisation of the strain–displacement relations and the use of cubic Hermite shape functions for the transverse displacements $v(s)$ and $w(s)$ and their slopes.

The shaft material is assumed homogeneous, isotropic and linearly elastic with constant Young's modulus E and density ρ. Internal damping is neglected. For each element, the cross–sectional area A, second moment of area I, and polar moment of inertia I_p are taken as constant, and warping is not considered. The translational mass matrix follows from the distributed mass per unit length ρA, while the rotary inertia matrix is derived from the polar and diametral moments of inertia of the cross–section. The gyroscopic matrix is obtained from the distributed polar inertia I_p assuming a constant spin speed Ω, retaining only the linear terms proportional to Ω. The bending stiffness matrix is computed from EI and the second derivatives of the Hermite shape functions; axial, torsional and shear deformations are not included in the lateral model. Finally, since no boundary constraints are applied at the shaft ends, the free–free model contains rigid–body modes with eigenfrequencies close to zero.

2.2 Mathematical Model

The lateral deformation of each shaft element is described using the cubic Hermite form functions, which interpolate both the transverse displacements and the corresponding rotations at the element boundaries. For an element of length l, the shape functions for the displacement field and their associated

rotational counterparts are given by

$$\begin{aligned}
\psi_1(s) &= 1 - \frac{3}{l^2}s^2 + \frac{2}{l^3}s^3 \quad \text{and} \quad \theta_1(s) = -\frac{6}{l^2}s + \frac{6}{l^3}s^2 \\
\psi_2(s) &= s - \frac{2}{l}s^2 + \frac{1}{l^2}s^3 \quad \text{and} \quad \theta_1(s) = 1 - \frac{4}{l}s + \frac{3}{l^2}s^2 \\
\psi_3(s) &= 2\frac{s^2}{l^2} - 2\frac{s^3}{l^3} \quad \text{and} \quad \theta_3(s) = \frac{6}{l^2}s - \frac{6}{l^3}s^2 \\
\psi_4(s) &= -\frac{s^2}{l} + \frac{s^3}{l^2} \quad \text{and} \quad \theta_4(s) = -\frac{2}{l}s + \frac{3}{l^2}s^2
\end{aligned} \tag{1}$$

These functions ensure the continuity of the displacement field, which is required for Euler–Bernoulli beam theory. Using these functions, the transverse displacements in the y- and z-directions can be written as

$$\begin{Bmatrix} V(s,t) \\ W(s,t) \end{Bmatrix} = \Psi(s)\,\mathbf{q}^e(t), \qquad \begin{Bmatrix} \beta(s,t) \\ \Gamma(s,t) \end{Bmatrix} = \Theta(s)\,\mathbf{q}^e(t) \tag{2}$$

where $\Psi(s)$ collects the displacement shape functions and $\mathbf{q}^e(t)$ contains the eight nodal degrees of freedom of the element. With $\Theta(s)$ containing the derivatives of the displacement shape functions.

The interpolation matrices take the form

$$\Psi(s) = \begin{bmatrix} \psi_1(s) & 0 & 0 & \psi_2(s) & 0 & 0 & \psi_3(s) & 0 & 0 & \psi_4(s) \\ 0 & \psi_1(s) & -\psi_2(s) & 0 & \psi_3(s) & -\psi_4(s) & & & & \end{bmatrix} \tag{3}$$

$$\Theta(s) = \begin{bmatrix} \Theta_\beta(s) \\ \Theta_\Gamma(s) \end{bmatrix} = \begin{bmatrix} 0 & -\theta_1(s) & 0 & \theta_2(s) & 0 & -\theta_3(s) & 0 & \theta_4(s) \\ \theta_1(s) & 0 & \theta_2(s) & 0 & \theta_3(s) & 0 & \theta_4(s) & 0 \end{bmatrix} \tag{4}$$

and the nodal displacement vector is

$$\mathbf{q}^e(t) = \left[q_1(t), q_2(t), q_3(t), q_4(t), q_5(t), q_6(t), q_7(t), q_8(t)\right]^T \tag{5}$$

Element Matrices Using the above interpolation fields, the element mass, gyroscopic and stiffness matrices follow directly from the kinetic and potential energy expressions of the Euler–Bernoulli beam.

Translational mass matrix. The contribution from the distributed mass $\mu = \rho A$ is obtained from

$$M_T^e = \int_0^l \mu\,\Psi^T(s)\,\Psi(s)\,ds \tag{6}$$

Rotational mass matrix. The rotary inertia associated with bending is included through

$$M_R^e = \int_0^l I_d\,\Theta^T(s)\,\Theta(s)\,ds \tag{7}$$

where I_d is the area moment of inertia of the cross-section.

Gyroscopic matrix. The gyroscopic coupling arises from the polar mass moment of inertia I_p and the rotation of the shaft with angular velocity Ω. The element gyroscopic matrix is

$$G^e = \int_0^l I_p\,\Theta_\Gamma^T(s)\,\Theta_\beta(s)\,ds \tag{8}$$

Bending stiffness matrix. The strain energy of bending leads to the stiffness matrix

$$K_B^e = \int_0^l EI\, \Psi''^T(s)\, \Psi''(s)\, ds \tag{9}$$

where EI is the bending stiffness of the shaft.

Element Equation of Motion Combining all contributions, the equation of motion for a single shaft element becomes

$$\left(M_T^e + M_R^e\right) \ddot{q}^e \; - \; \Omega\, G^e \dot{q}^e \; + \; K_B^e q^e \; = \; Q^e \tag{10}$$

where Q^e contains the external Loads acting on the element nodes.

By assembling the element matrices according to the mesh of the shaft, the resulting global equation of motion can be written as

$$M\ddot{q} - \Omega G\dot{q} + Kq = Q, \tag{11}$$

where M, G and K are the assembled global matrices. The gyroscopic and stiffness matrices are obtained from the element contributions M_T^e, M_R^e, G^e and K_B^e.

The different discretization levels correspond to increasingly finer finite element meshes of the same shaft geometry, as summarized 2.

2.3 Shaft Segmentation and Discretization

Since the Shaft varies in thickness it is necessary to differentiate it into different segments. In table 1 those segments with their corresponding thickness and lenght are presented.

Table 1: Shaft segment-coordinates thickness and lenghts

Typ.	x_{init}	n_1	n_2	n_3	n_4	n_5	n_6	n_7	n_8	n_9	n_{10}	n_{11}	n_{11}
x_{end}	0	52	75	110	700	735	855	885	910	1085	1125	1135	1150
l_i	-	52	23	35	590	35	120	30	25	175	40	10	15
d_i	-	40	50	70	90	90	99.6	90	90	70	65	65	14

Since the lengths of the shaft segments vary significantly e.g. from a minimum of 10 mm to a maximum of 590 mm, a direct use of the geometric segmentation as the finite element discretisation leads to numerical difficulties. The large variation in element length produces unbalanced element matrices, since the bending stiffness term $\frac{12EI}{l^3}$ becomes extremely sensitive to small values of l. As a consequence, the global stiffness and mass matrices differ by several orders of magnitude, and the resulting eigenvalue problem cannot be solved reliably.

To avoid this issue, the geometric segmentation is used only to consider the correct shaft diameters, while each segment is further subdivided into elements of approximately constant length. The investigated discretisation levels are summarised in table 2.

For discretisation levels up to level 4, the default eigenvalue solver implemented in the provided FEM script *flexible-rotor-modal-DTU.m* performs well. However, for the fifth discretisation level, where the maximum element length $l_{\text{max,elem}} = 10$ mm becomes very small, the stiffness matrix contains entries of the order $l^{-3} = 10^{-6}$, which dominate the mass matrix. This results in unrealistically high natural frequencies.

To overcome this problem, the `eigs` solver was used with the option `'smallestabs'`, which extracts the eigenvalues with the smallest absolute magnitude and therefore avoids convergence towards the artificially large eigenvalues caused by the extremely stiff elements. Further adjustments were needed for the seventh discretization level, where elements of 1 mm length produced very large stiffness

matrix entries. To restore numerical balance, the dummy bearing stiffness was increased by a factor of ten, ensuring comparable matrix contributions from both the shaft and bearing terms. As shown in Section 4.1, varying the bearing stiffness over a wide range has little effect on the higher natural frequencies, which confirms that this adjustment does not influence the results.

Table 2: Discretisation level of the shaft, for level 1 to 4 eig solver used, for level 5 to 7 eigs solver implemented. For level 7 the dummy stiffness of the bearings had to be increased from previously 100 N/mm² to 1000 N/mm² the reduce the unbalance within the stiffness matrices.

Discr.	l_{min} [mm]	l_{max} [mm]	n_{elem}	ω_1 [Hz]	ω_2 [Hz]	ω_3 [Hz]	ω_4 [Hz]	t [s]
1	10	98.33	21	384.19	958.60	1707.84	2770.88	0.034
2	10	50	31	384.25	958.54	1708.27	2765.02	0.075
3	10	40	36	384.26	958.58	1708.21	2763.87	0.106
4	10	20	63	382.43	960.9	1700.16	2788.64	0.791
5	5.75	10	120	384.17	958.24	1707.57	2763.84	0.29
6	3.83	5	232	384.17	958.24	1707.57	2763.83	0.789
7	0.96	1	1151	384.14	958.17	1707.57	2763.78	21.5

Figure 1 shows the first four flexible bending mode shapes of the shaft in a free-free condition. The modes follow the expected pattern for a free-free beam: the first mode has three peaks and two nodal points, the second four peaks and three nodes, and so on, with each successive mode adding one additional peak and one additional node. This is consistent with the theoretical behaviour of higher bending modes of a slender beam.

The non-zero displacement amplitudes at the bearing node locations confirm that the dummy bearing stiffness of 100 MPa is sufficiently small to leave the shaft essentially unconstrained, reproducing the free-free boundary condition as intended.

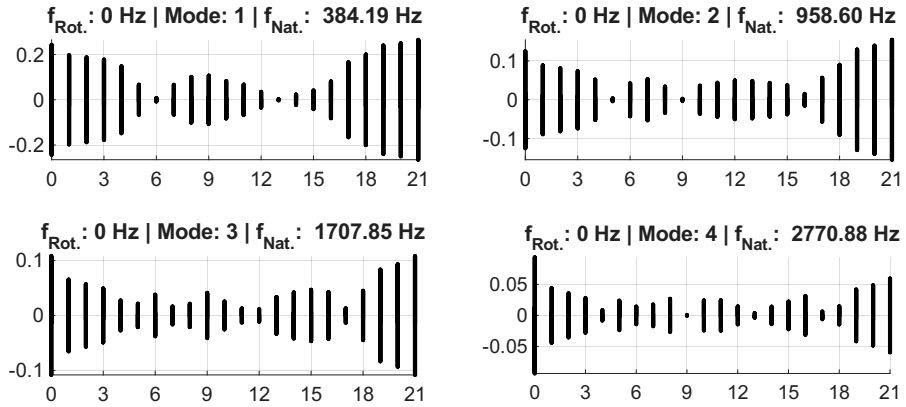


Figure 1: The first four not rigid Modeshapes of System I, only the shaft in a free-free condition

2.4 Validation

The experimental reference frequencies are obtained from the modal testing of the shaft in an almost free–free condition and excited laterally using an instrumented impact hammer at several nodes (4, 8, and 18). The resulting acceleration response is measured at node 10 using a piezoelectric accelerometer. Each time signal was subsequently processed using a Fast Fourier Transform (FFT), identifying the dominant resonance peaks.

Across all excitation locations, four consistent natural frequencies are observed in the range 0–3000 Hz, namely 388 Hz, 930 Hz, 1689 Hz, and 2624 Hz. Since these peaks appear independently of the excitation point and are well separated from background noise, they can be considered reliable experimental estimates of the first four bending modes of the free–free shaft. These frequencies serve as

the reference values for validating the finite element model. The relative deviation between the theoretical and experimental natural frequencies is defined as:

$$\Delta[\%] = \frac{\omega_{\text{th}} - \omega_{\text{exp}}}{\omega_{\text{exp}}} \cdot 100\,\% \tag{12}$$

where ω_{th} is the theoretically predicted frequency and ω_{exp} the corresponding experimentally measured value. The results are summarised in Table 3.

Table 3: Comparison of the theoretical natural frequencies with the experimental ones ω_1 = 388 Hz, ω_2 = 930 Hz, ω_3 = 1689 Hz, ω_4 = 2624 Hz

Discr.	ω_1 [Hz]	Δ_1 [%]	ω_2 [Hz]	Δ_2 [%]	ω_3 [Hz]	Δ_3 [%]	ω_4 [Hz]	Δ_4 [%]	t [s]
1	384.19	-0.98	958.6	3.08	1707.84	1.12	2770.88	5.6	0.0338
2	384.25	-0.97	958.54	3.07	1708.27	1.14	2765.02	5.37	0.0748
3	384.26	-0.96	958.58	3.07	1708.21	1.14	2763.87	5.33	0.1059
4	382.43	-1.44	960.9	3.32	1700.16	0.66	2788.64	6.27	0.7909
5	384.17	-0.99	958.24	3.04	1707.57	1.10	2763.84	5.33	0.2899
6	384.17	-0.99	958.24	3.04	1707.57	1.10	2763.83	5.33	0.7882
7	384.14	-0.99	958.17	3.03	1707.57	1.10	2763.78	5.33	21.4952

The discretisation levels 5, 6, and 7 are essentially identical. This behaviour is expected, since from level 5 onward the maximum element length is already sufficiently small (less than 10 mm), such that further mesh refinement does not provide additional accuracy for the first bending modes. Moreover, the use of the `eigs` solver with the `smallestabs` option finds numerically stable results once the mesh has reached this resolution, which explains the convergence plateau across the finest discretisation levels.

Discretisation level 4 shows slightly larger deviations compared to other refinement levels. This might be due to numerical treatment of the eigenvalue problem. For levels 1–3, the standard `eig` solver can still handle the system matrices despite the heterogenous element lengths. From level 4 onward, however, the element lengths eventually become small enough that the stiffness matrix starts dominating the mass matrix, leading to an increasingly badly-conditioned eigenvalue problem. Consequently, the standard solver starts becoming unreliable indicating the need to change to the `eigs` solver for the following discretization levels.

2.5 Adjustment

For the adjustement of the shaft it should be considered that the deviation of the first natural frequencies is already quite low as shown in table 3. For that reason adjusting the youngsmodulus E or the density ρ of the shaft in order to reduce the deviation would shift all natural frequencies equally not really causing an improvement for the higher natural frequencies. Because Euler-Bernoulli beam elements, which neglect shear deformation, were used for the shaft, the model is structually stiff at higher natural frequencies. To numerically compensate for this lack of shear deformation at shorter bending wavelengths, the rotary inertia matrix was scaled with a factor of 2. This adjustment is aimed to correct the higher natural frequencies rather than the first bending frequencies. Table 8

Table 4: Comparison of the theoretical natural frequencies with the experimental ones ω_1 = 388 Hz, ω_2 = 930 Hz, ω_3 = 1689 Hz, ω_4 = 2624 Hz. The deviation Δ_i has been calculated according to equation 12.

Discr.	ω_1 [Hz]	Δ_1 [%]	ω_2 [Hz]	Δ_2 [%]	ω_3 [Hz]	Δ_3 [%]	ω_4 [Hz]	Δ_4 [%]	t [s]
1	380.56	-1.92	942.76	1.37	1657.04	-1.89	2646.56	1.24	0.0409
2	380.58	-1.91	942.46	1.34	1656.57	-1.92	2652.85	1.1	0.0794
3	380.58	-1.91	942.48	1.34	1656.6	-1.92	2652.93	1.10	0.1371
4	382.58	-1.4	946.441	1.77	1678.69	-0.61	2673.16	1.87	0.7704
5	380.5	-1.93	942.13	1.3	1655.59	-1.98	2649.75	0.98	0.3155
6	380.5	-1.93	942.13	1.3	1655.59	-1.98	2649.75	0.98	0.8555
7	380.5	-1.93	942.13	1.3	1655.59	-1.98	2649.75	0.98	22.5612

2.6 Conclusion

The mesh refinement has only a minor influence on the natural frequencies, but it is essential for avoiding unbalanced system matrices in the eigenvalue solvers (eig and eigs). Since some shaft segments only have a length of 15 mm, while others reach up to 590 mm, the large variation in element length affects the stiffness matrix, where the term $12EI/l^3$ becomes highly sensitive to small l. Refining the mesh therefore serves mainly to regularise the element sizes and improve numerical conditioning.

Due to no significant improvement in the accuracy of the natural frequency prediction and since the computational time increases noticeably (see table 2), it is more efficient to select the lowest refinement level that still ensures stable matrix conditioning. In this case, the mesh shown in figure 2 with a maximum element size of 98.33 mm provides a good balance between numerical stability and computational cost.

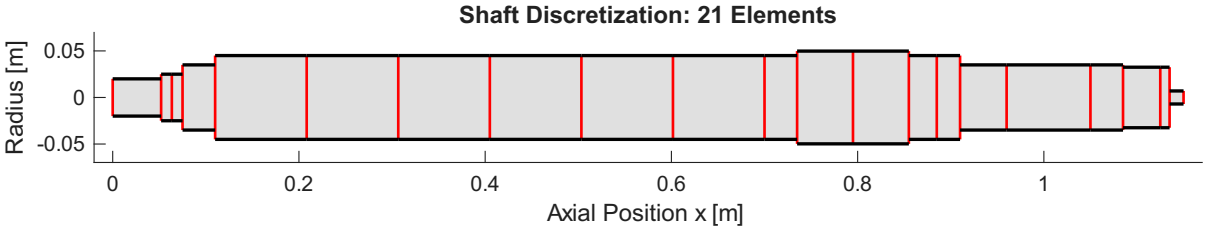


Figure 2: Chosen shaft discretization for the following tasks with a maximum element size of 98.33 mm

3 Modelling the Lateral Dynamics of Flexible Shaft and Discs

3.1 Modelling Assumptions for the Rigid Disc

The rigid disc finite element is built under the assumption that the disc undergoes no elastic deformation and can therefore be treated as a perfectly rigid body. It is assumed to be axisymmetric and made of a homogeneous, isotropic material with constant density, so that the mass distribution is symmetric with respect to the shaft axis and the diametral mass moments of inertia about the two lateral axes are equal. Small lateral displacements and small rotations are assumed, consistent with the linearised kinematics adopted for the shaft elements. Finally, the disc is taken to be perfectly concentric and rigidly attached to the shaft node, meaning that the disc degrees of freedom coincide directly with those of the corresponding shaft node.

3.2 Mathematical Model of the Disc

The rigid disc is modeled as a non-deformable body attached to a shaft node. Its generalized coordinates are the lateral translations and rotations

$$q^d = \begin{bmatrix} V & W & \beta & \Gamma \end{bmatrix}^\mathsf{T}. \tag{13}$$

where V and W are the lateral displacements in the two transverse directions, and β and Γ are the corresponding bending rotation angles about the respective transverse axes.

The equation of motion is obtained from Lagrange's equation

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial E_{\mathrm{kin}}}{\partial \dot{q}_i}\right) - \frac{\partial E_{\mathrm{kin}}}{\partial q_i} + \frac{\partial E_{\mathrm{pot}}}{\partial q_i} = Q_i, \qquad i = 1, \ldots, 4. \tag{14}$$

The kinetic energy of the disc consists of translational and rotational parts:

$$E_{\mathrm{kin}} = \frac{1}{2} m_D (\dot{V}^2 + \dot{W}^2) + \frac{1}{2} I_D (\dot{\beta}^2 + \dot{\Gamma}^2) + I_P (\frac{1}{2} I_P \dot{\phi}^2 - \dot{\phi}\beta\dot{\Gamma}) \tag{15}$$

where m_D is the disc mass, I_D the diametral moment of inertia, I_P the polar moment of inertia and ϕ the angular velocity. The rigid disc has no elastic deformation, therefore $E_{\mathrm{pot}} = 0$.

Substituting the kinetic energy into Lagrange's equation results in the finite element equation of the rigid disc:

$$M^d \ddot{q}^d - \Omega G^d \dot{q}^d = Q^d, \tag{16}$$

with the mass matrix M^d and the gyroscopic matrix G^d

$$M^d = \begin{bmatrix} m_D & 0 & 0 & 0 \\ 0 & m_D & 0 & 0 \\ 0 & 0 & I_D & 0 \\ 0 & 0 & 0 & I_D \end{bmatrix}, \quad G^d = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -I_P \\ 0 & 0 & I_P & 0 \end{bmatrix}. \tag{17}$$

The vector Q^d contains external loads acting on the disc. The matrices M^d and G^d are finally assembled into the global finite element system at the corresponding shaft node.

3.3 Modelling and Refinement of the Rigid Discs

To accurately determine the inertial properties of the discs, a detailed geometric assessment is required. In the initial validation, the disc was simplified as a single homogeneous hollow cylinder with a constant external and internal diameter[cite: 140]. To improve the model's accuracy and minimize the deviations in natural frequencies, the disc is now modeled as a series of axial segments.

3.3.1 Geometric Definition

The disc geometry features several steps in both the internal bore and the external contour. Table 5 summarizes the measurements of the discs used for the calculation of the total mass (m_D), the polar (I_P) and diametral (I_D) mass moments of inertia:

Table 5: Geometry of the segmented contour of the two discs

	Disc 1 mounted to shaft [mm]							Disc 1 & 2 mounted to shaft [mm]						
Segments.	1	2	3	4	5	6	7	1	2	3	4	5	6	7
$D_{in,i}$	70	90	110					70	90	110				
$l_{in,i}$	55	20	25					35	20	25				
$D_{out,j}$	295	271	295	271	295	271	295	295	271	295	271	295	271	295
$l_{out,j}$	10	10	25	10	25	10	10	10	10	25	10	25	10	10

The discs are mounted on the shaft at the axial positions $x_{D1} = 995\,\mathrm{mm}$ and $x_{D2} = 1085\,\mathrm{mm}$.

3.3.2 Calculation of Inertial Parameters

Instead of the global simplification, the total mass m_D, as well as the polar (I_P) and diametral (I_D) mass moments of inertia, are calculated by the superposition of individual cylindrical segments. Assuming a constant density ρ, the mass is determined by the difference between the sum of the external and internal volumes:

$$m_D = \rho \left(\sum_{j=1}^{7} \frac{\pi}{4} D_{out,j}^2 \cdot l_{out,j} - \sum_{i=1}^{3} \frac{\pi}{4} D_{in,i}^2 \cdot l_{in,i} \right) \tag{18}$$

The polar mass moment of inertia I_P is determined similarly as the sum of the polar moments of the individual segments:

$$I_P = \frac{\rho \pi}{32} \left(\sum_{j=1}^{7} D_{out,j}^4 \cdot l_{out,j} - \sum_{i=1}^{3} D_{in,i}^4 \cdot l_{in,i} \right) \tag{19}$$

For the diametral mass moment of inertia I_D, the axial distribution of the segments must be taken into account. Since the segments are located at different positions along the shaft axis, the Parallel Axis Theorem (Steiner's Theorem) is applied. First, the axial center of gravity z_{cg} of the entire disc is calculated:

$$z_{cg} = \frac{1}{m_D} \left(\sum_{j=1}^{7} m_{out,j} \cdot z_{out,j} - \sum_{i=1}^{3} m_{in,i} \cdot z_{in,i} \right) \tag{20}$$

where $z_{out,j}$ and $z_{in,i}$ represent the local axial center coordinates of each respective segment. The total diametral moment of inertia I_D is then determined by summing the local inertias of each segment and their corresponding Steiner terms ($m \cdot d^2$):

$$I_D = \sum_{j=1}^{7} \left[I_{D,out,j} + m_{out,j} (z_{out,j} - z_{cg})^2 \right] - \sum_{i=1}^{3} \left[I_{D,in,i} + m_{in,i} (z_{in,i} - z_{cg})^2 \right] \tag{21}$$

where the local diametral inertia for each segment is given by the standard cylinder formula:

$$I_{D,seg} = m_{seg} \left(\frac{1}{4} \left(\frac{D_{seg}}{2} \right)^2 + \frac{1}{12} l_{seg}^2 \right) \tag{22}$$

3.4 Validation

The experimental natural frequencies used for validation were obtained from the modal tests shown in the Appendix 7.1 in Figure 20 and Figure 21. In both configurations (Disc 1 solely and Disc 1+2), the shaft was suspended in an almost free–free condition and excited laterally using an instrumented impact hammer at several nodes. The lateral acceleration response was measured at node 10 with a piezoelectric accelerometer.

The identified natural frequencies are consistent across different excitation locations, which indicates that the measured peaks correspond to global bending modes of the shaft–disc system rather than local artefacts. Furthermore, the amplitude of the peaks varies significantly with the excitation node, reflecting the mode shapes, modes with a node close to the excitation point are weakly excited, whereas modes with large local curvature appear more clearly. This explains, for example, why the additional peak at 1995 Hz appears only for the Disc 1 configuration when exciting at node 6. It is also to consider that the experimental setup is not perfectly free–free, since the shaft is suspended by soft elastic bands, and the accelerometer mass introduces a small local mass, however those contributions of marginal magnitude.

When comparing the theoretical and experimental results in Table 8, the configuration with a single disc shows moderate deviations for all modes, with the first mode having the smallest error (-0.68 %). In contrast, the system with two discs has larger discrepancies, particularly for the third mode, where the deviation reaches 15.57 %. This mode behaves as a clear outlier compared to the remaining estimates and indicates that the model does not capture the dynamic behaviour of the two-disc configuration equally well. The first and second modes also show increased deviations once the second disc is added, indicating that the additional mass and inertia distribution introduce modelling inaccuracies.

Table 6: Comparison of the theoretical natural frequencies with the experimental ones Disc 1 with a thickness of 100 mm and Disc 2 with a thickness of 80 mm. The deviation Δ has been calculated according to equation 12.

	Disc 1 mounted to shaft			Disc 1 & 2 mounted to shaft		
Symbol	$\omega_{th.}$ [Hz]	$\omega_{\text{exp.}}$ [Hz]	Δ [%]	$\omega_{th.}$ [Hz]	$\omega_{\text{exp.}}$ [Hz]	Δ [%]
ω_1	298.96	301	-0.68	234.75	241	-2.59
ω_2	622.96	669	-6.88	493.23	498	-0.96
ω_3	1318.32	1348	-2.20	817.07	707	15.57
ω_4	1840.2	1995	-7.76	1328.96	1250	6.32
ω_5	2482.5	2383	4.18	2405.03	2350	2.34

Figure 3 shows the first five flexible mode shapes of System II (one disc), and Figure 4 shows the corresponding modes of System III (two discs).

Comparing the two systems, the first two mode shapes are nearly identical, and Mode 3 is also very similar, although a slight shift in the nodal points is already visible. Mode 5 is again comparable between the two systems, apart from a difference in amplitude.

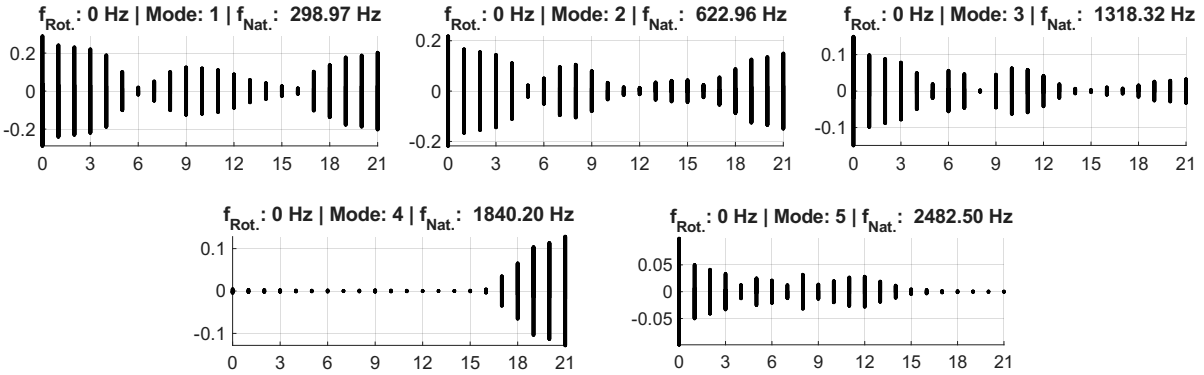


Figure 3: Modeshapes of System I only the disc of 100 mm thickness implemented

Mode 4, however, differs significantly. In System II the deformation is concentrated near the disc location, whereas in System III it shows a smoother bending pattern along the shaft. This can be explained by the addition of the second disc, which introduces a new inertia into the system. The natural frequency of this local disc-dominated mode falls close to the fourth global bending frequency, causing a mode veering so the two modes interact and exchange their shapes. As a result, what appears as the fourth mode in System II is strongly influenced by the local motion of the disc, while the second disc in System III redistributes this effect across the shaft.

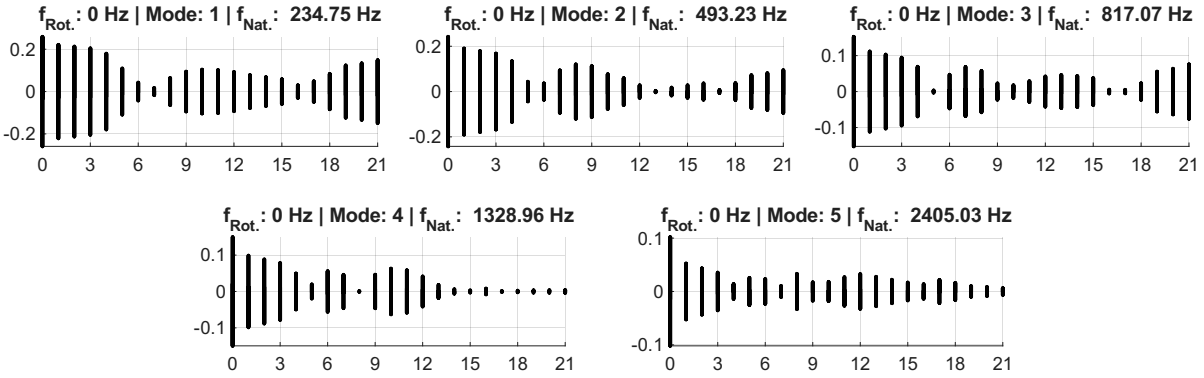


Figure 4: Modeshapes of System 2 both discs mounted to the shaft

3.5 Adjustment

The comparison between the theoretical and experimental natural frequencies in Table 8 shows that the unadjusted model overestimates the eigenfrequencies for both System II (only Disc 1 mounted) and System III (both discs mounted). This indicates that the mathematical model is effectively "too stiff" or "too light" compared to the real structure.

System II For System II (shaft with Disc 1), the unadjusted model shows deviations between $-7.76\,\%$ and $4.18\,\%$, with the largest error occurring in the fourth bending mode. These lower modes are mainly governed by the global mass distribution of the rotor. To increase the theoretical frequencies and reduce the systematic underestimation, the effective density of the shaft was scaled to $\rho = 0.85$ (see Table 7), compensating for an overestimation of the mass due to material variability and eventual manufacturing tolerances. In addition, the diametral moment of inertia of the disc was reduced by $15\,\%$ to reflect a non-ideal, not perfectly rigid mounting on the shaft.

After applying these adjustments, the deviations for System II fall below $5\,\%$ for all modes (Table 8). The correction produces a lighter rotor with reduced rotational inertia, which increases the natural frequencies and aligns the model with the experimental data. The fifth mode remains largely unaffected, as it was already slightly overestimated in the unadjusted model and is dominated by local high-frequency dynamics that are less sensitive to global mass scaling, which is why the density has a minor influence on the fifth mode.

System III When Disc 2 is added (System III), the unadjusted model shows noticeably larger deviations than in System II. The first two natural frequencies are underestimated, while the upper three modes are overestimated. The largest errors occur in the third and fourth modes ($15.57\,\%$ and $6.32\,\%$), indicating that the simplified disc representation does not capture the combined mass and inertia effects of the two-disc configuration accurately.

A direct correction by increasing the mass of the Discs does not work, because the lower modes are more sensitive to global mass changes than the higher ones. Reducing the third mode by making the disc heavier would therefore shift the first two modes too far downward. In contrast, the rotational mass matrix M_R mainly affects the higher modes. Increasing the diametral moment of inertia lowers the natural frequencies more strongly in the upper range. However, applying this correction alone (factor 1.5) reduces the lower modes too much, so the density of the shaft and discs must be reduced to compensate and raise the lower frequencies again.

The final adjustment therefore combines both effects, reducing the density of the shaft and applying an additional mass reduction directly to the discs. Lowering the density alone would also reduce the moments of inertia, but these need to be increased to correct the higher modes. For this reason, part of the mass reduction is applied specifically to the disc masses, which lightens the system and raises the first two natural frequencies. At the same time, the diametral moments of inertia of Disc 1 and Disc 2

are increased by 48 % and 25 %, respectively, to lower the higher modes and bring them into the correct range. The resulting parameter set in Table 7 brings all five natural frequencies of System III within the desired deviation range.

Table 7: Adjusted Parameters

System	ρ	$m_{\text{Disc,1}}$	$m_{\text{Disc,2}}$	I_{d1}	I_{d2}	M_R
II	0.85	-	-	0.85	-	-
III	0.9	0.8	0.8	1.48	1.25	1.5

Table 8: From adjustment resulting natural frequencies and deviation from the experiments for System II (only Disc 1 with 100 mm thickness) and for System III (both disc mounted). The deviation Δ has been calculated according to equation 12.

	Disc 1 mounted to shaft			Disc 1 & 2 mounted to shaft		
Sign	$\omega_{adj.}$ [Hz]	$\omega_{\text{exp.}}$ [Hz]	Δ_{adj} [%]	$\omega_{adj.}$ [Hz]	$\omega_{\text{exp.}}$ [Hz]	Δ_{adj} [%]
ω_1	313.39	301	4.12	229.1	241	4.94
ω_2	668.95	669	-0.01	490.31	498	1.54
ω_3	1335.36	1348	-0.94	742.11	707	4.97
ω_4	1900.93	1995	-4.72	1304.91	1250	4.39
ω_5	2494.16	2383	4.66	2408.53	2350	2.49

4 Modelling the Ball Bearing mounted to the Shaft

4.1 Influence of Varying Bearing Stiffness

4.1.1 System I - Only the Shaft

Figure 5 shows the natural frequencies of System I for a bearing stiffness sweep from 10^1 to 10^{11} N/m. For low stiffness values (10^1–10^8), the mode shapes change only slightly. At these levels, the bearings at nodes 3 and 12 are too soft to influence the deformation pattern, and the shaft behaves almost as a free–free system. From 10^8 upward, a clear change becomes visible, especially in the first two modes, whose trends shift significantly as the bearings begin to constrain the motion. At 10^{11} N/m the bearings act effectively rigid, and the nodal displacements at the bearing locations are fully suppressed.

The higher modes are much less affected by the bearing stiffness. Their shorter wavelengths and multiple internal nodes mean that most deformation occurs within the shaft spans rather than at the supports. As a result, the bearings contribute only a small fraction of the modal strain energy, so increasing their stiffness has little influence on the corresponding natural frequencies and modeshapes.

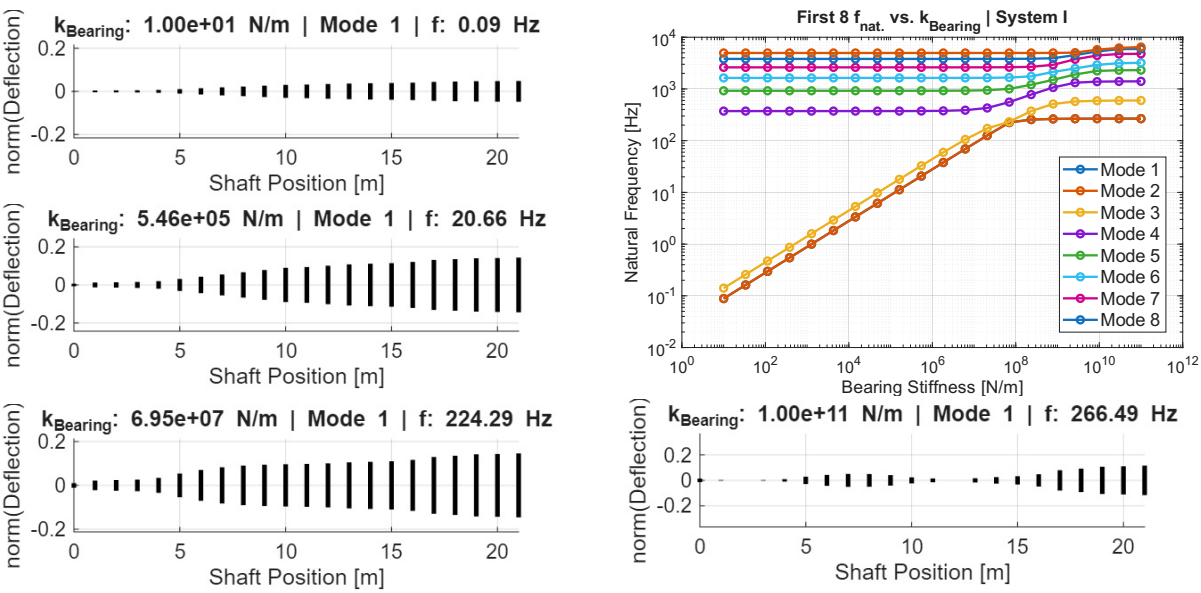


Figure 5: First 8 Natural Frequencies of System I vs Bearing Stiffness with Corresponding Modeshape Plots of the First Mode

4.1.2 System II – Shaft with One Disc

Compared to the stiffness sweep of System I (Figure 5), the first two natural frequencies in Figure 6 move less closely together. This shows that the mounted disc has a clear influence on the mode shapes. The plotted mode shapes represent points along the natural-frequency curves for different bearing stiffness values.

The disc is mounted at node 4, which becomes visible in the mode shapes at very low bearing stiffness. In this range, the inertia of the disc constrains the local motion of the shaft, and the nodal displacement at the disc position is almost zero. As the bearing stiffness increases, the disc begins to participate more in the deformation of the mode shape, which can be seen around 220.71 Hz. At the highest stiffness level (10^{11} N/m), the bearings behave as rigid supports, and the mode shapes again resemble those of a fixed–fixed shaft, similar to System I.

The higher modes are only weakly affected by the bearing stiffness. Their shorter wavelengths and multiple internal nodes mean that most deformation occurs between the bearings rather than at the support locations. Because the bearings contribute only a small part of the modal strain energy in these modes, increasing the bearing stiffness has little influence on their shape or frequency.

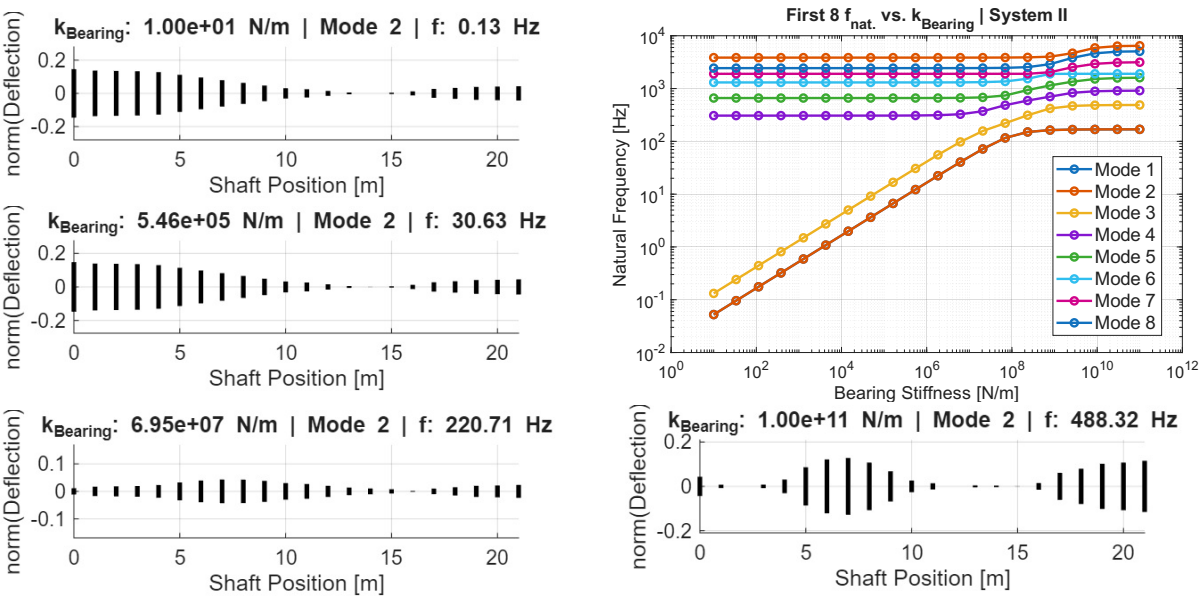


Figure 6: First 8 Natural Frequencies of System II vs Bearing Stiffness with Corresponding Modeshape Plots of the Second Mode

4.2 Critical Speed of System II (Two Ball Bearings)

The Campbell diagram in Figure 14 shows the natural frequencies of all three systems as a function of rotor speed, together with the 1X synchronous excitation line. At zero speed, the first natural frequencies are approximately 270 Hz, 170 Hz, and 110 Hz for Systems I, II, and III respectively. The significant drop in natural frequency with each added disc reflects the increase in mass, which lowers the resonance frequencies of the system. The second natural frequencies follow the same trend, starting at approximately 600 Hz, 485 Hz, and 400 Hz for the three systems.

The first critical speeds, defined by the intersection of the lower frequency branch with the 1X synchronous line, are approximately 260 Hz, 150 Hz, and 100 Hz for Systems I, II, and III respectively. This shows that adding disc mass strongly reduces the critical speed, which has direct implications for the operational range of the machine.

Regarding the gyroscopic splitting of the frequency branches, System I shows almost no divergence between the forward and backward whirl frequencies, since the shaft alone contributes only marginal polar inertia. Systems II and III diverge at a similar rate for the first mode, suggesting that the second disc does not significantly increase the gyroscopic behaviour at this frequency. This makes physical sense, as the gyroscopic effect scales with the polar moment of inertia, and the second disc is smaller than the first. For the second mode, however, the divergence between Systems II and III becomes more pronounced, indicating that the polar inertia of the second disc has a greater relative influence on higher modes.

The first mode shapes of all three systems at their respective critical speeds are shown in Figure 15. All three systems have the same general shape with two peaks along the shaft. The natural frequency at the critical speed is 42 Hz, 23 Hz, and 16 Hz for Systems I, II, and III respectively, again reflecting the increased mass of the added discs.

The main difference between the mode shapes lies in the amplitude distribution. System I shows the highest amplitude in the shaft section between the bearings, while this amplitude decreases for Systems II and III. Near the disc position and the second bearing, the amplitudes are comparable across all three systems.

The added disc mass increases the rotating mass of the system, which primarily affects the lower modes, as visible by the decrease in natural frequency of each first modeshape with every disc added. Since the discs are mounted very close to the bearing positions, they are located near enforced nodes of

the modeshape, meaning their contribution to amplitude change is small, particularly for the first mode. As a result, the deformation near the disc and bearing locations remains largely unaffected, while the relative flexibility of the shaft mid-section is reduced. As a result, the deformation near the disc and bearing locations remains largely unaffected, while the increased rotating mass introduced with each disc stiffens the dynamic response of the shaft mid-section relative to the rest of the structure. Consequently, adding discs lowers the critical speed and redistributes the vibration amplitude along the shaft. The gyroscopic frequency splitting becomes increasingly significant with greater disc inertia, particularly for higher modes where the polar inertia contribution of the discs is more influential relative to the shaft stiffness.

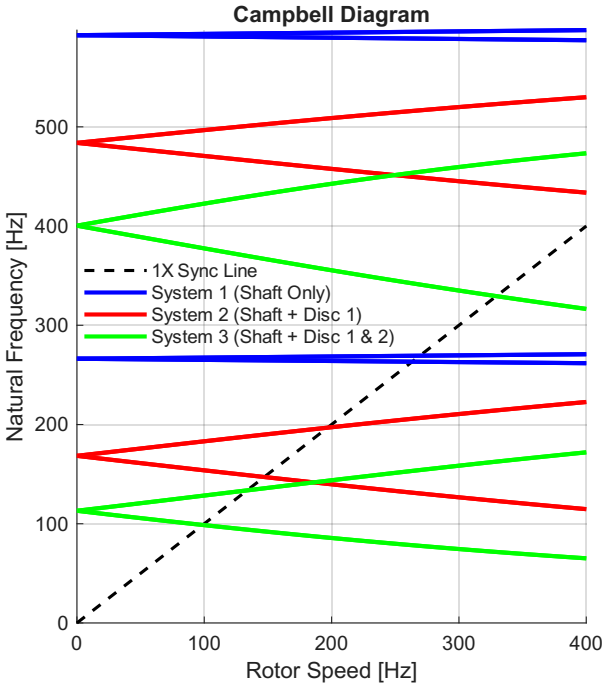


Figure 7: Campbell Diagramm

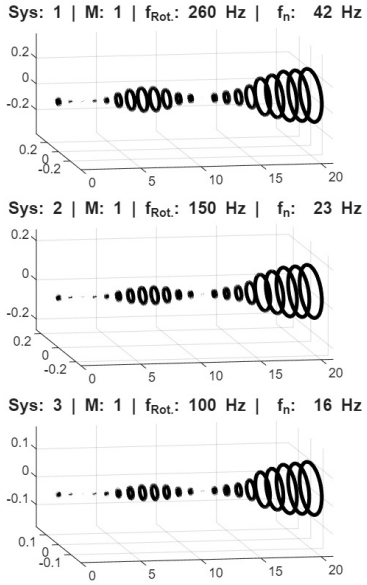


Figure 8: First Modeshapes of System I - III

4.3 Unbalance Response Using Two Ball Bearings

As shown in Figure 15, the vibration amplitude at the disc grows without bound at the first critical speed of approximately 150 Hz. This is a direct consequence of the system being purely undamped: with two ball bearings and no damping source, there is nothing to limit the resonance amplitude.

It is therefore not possible to satisfy the 30 µm design limit while crossing the critical speed. No finite unbalance can be defined for an undamped system at resonance, since the amplitude diverges for any non-zero unbalance. In reality, some damping is always present, but within this undamped ball bearing model the design requirement cannot be met.

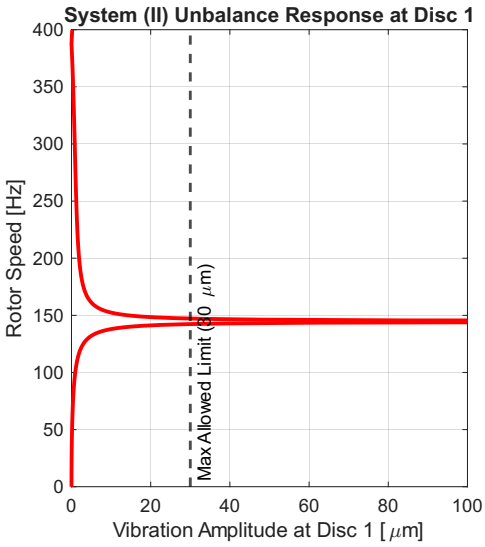


Figure 9: Unbalance response of System II

5 Modelling the Hydrodynamic Bearing mounted to the Shaft

The journal bearing data used in this project is derived from the classical Reynolds equation, which is not obtained directly from the full Navier–Stokes equations but from a sequence of simplifying assumptions that reduce the three-dimensional fluid problem to a two-dimensional thin-film lubrication model. The validity of the bearing tables therefore depends on how well these assumptions hold in practice.

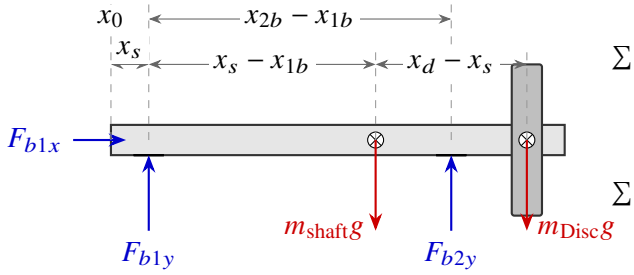
The lubricant is treated as a Newtonian fluid with constant viscosity, obeying a linear stress–strain-rate relationship, and the flow is assumed to be laminar, meaning that the Reynolds number in the oil film is sufficiently small for turbulence to be absent. The thin-film approximation requires that the film thickness h is much smaller than the bearing diameter D, so that pressure gradients across the film thickness can be neglected and pressure varies only in the axial and circumferential directions. Fluid inertia terms are dropped from the governing equations, leaving viscous forces as the dominant contribution to the flow behaviour.

The oil temperature is assumed constant throughout the film, so that viscosity does not vary inside the bearing. Both the shaft and bearing surfaces are treated as perfectly rigid, ignoring any elastohydrodynamic deformation. A no-slip condition is applied at both surfaces, meaning the fluid velocity equals the local surface velocity at the boundaries. The bearing is assumed to operate at steady-state or quasi-steady conditions, so that time derivatives of the pressure field are neglected. Cavitation is handled through the half-Sommerfeld condition, which prevents the pressure from falling below the lubricant vapour pressure. Finally, the journal and bearing axes are assumed to be perfectly parallel, excluding misalignment and tilting-pad effects from the model.

If any of these assumptions are violated under real operating conditions, the stiffness, damping, and equilibrium predictions obtained from the bearing tables may no longer be reliable.

5.1 Hydrodynamic Bearing - Static and Dynamic Properties

Taking moments about the first bearing and applying vertical force equilibrium as shown in the figure 10, the static loads acting on the two bearings are found to be $F_{b1y} = 63.94\,\text{N}$ and $F_{b2y} = 808.02\,\text{N}$. The large difference is expected given the overhung disc configuration, which places most of the weight on the bearing closest to the disc.



$$\sum M_{b1} = 0 \Rightarrow \quad F_{b2y}(x_{2b} - x_{1b}) - m_{\text{shaft}}g(x_s - x_{1b}) - m_{\text{Disc}}g(x_d - x_{1b}) = 0 \tag{23}$$

$$\sum F_y = 0 \Rightarrow \quad F_{b1y} + F_{b2y} - m_{\text{shaft}}g - m_{\text{Disc}}g = 0 \tag{24}$$

Figure 10: Free body diagram of the shaft–disc system with associated equilibrium equations. The center of gravity x_s is determined according to Equation 20.

Figure 11 shows the minimum oil film thickness as a function of rotor speed for both bearing types. The bearing with $L/D = 1.0$ builds up oil film thickness much faster, reaching its plateau of around 107 µm already below 50 Hz. The bearing with $L/D = 0.5$ increases more gradually, only reaching a similar plateau well above 100 Hz. Both curves level off because the eccentricity ratio approaches its minimum value at high speeds. Note that for both bearings the table data only covers speeds up to approximately 200 Hz, so the interpolated values beyond that range should be treated with caution. From this perspective, the bearing with $L/D = 1.0$ is preferable, as it provides a thicker oil film at low speeds, reducing the risk of metal-to-metal contact during startup and rundown when passing through the critical speed.

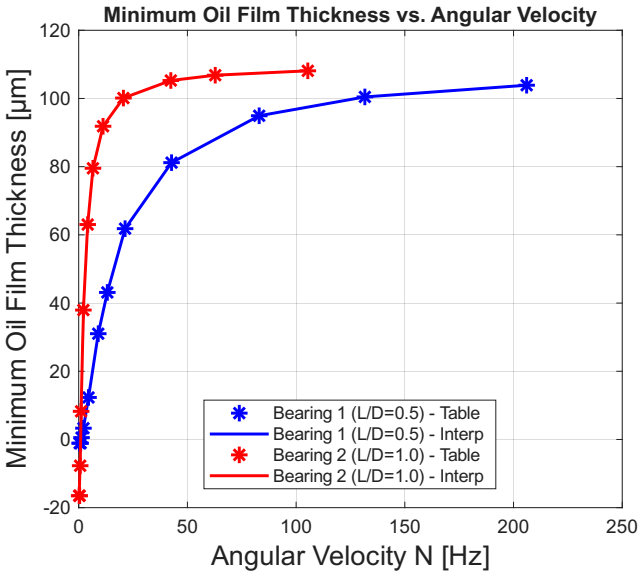


Figure 11: Minimum oil film thickness

Stiffness and Damping Coefficients The stiffness and damping coefficients for both bearing types are plotted as a function of angular velocity in Figures 12 and 13, respectively. The star symbols represent the original table values and the continuous lines the piecewise linear interpolation.

Regarding stiffness, the bearing with $L/D = 1.0$ shows a steep increase in all coefficients with rising angular velocity. The direct stiffness k_{xx} reaches values up to 2.5×10^9 N/m, while k_{xy} and k_{yy} also increase significantly. The cross-coupling term k_{yx} decreases strongly and becomes largely negative, indicating a destabilising coupling between the two lateral directions. In contrast, the $L/D = 0.5$ bearing shows a much flatter response across all stiffness coefficients, remaining well below the values of the $L/D = 1.0$ bearing throughout the entire speed range.

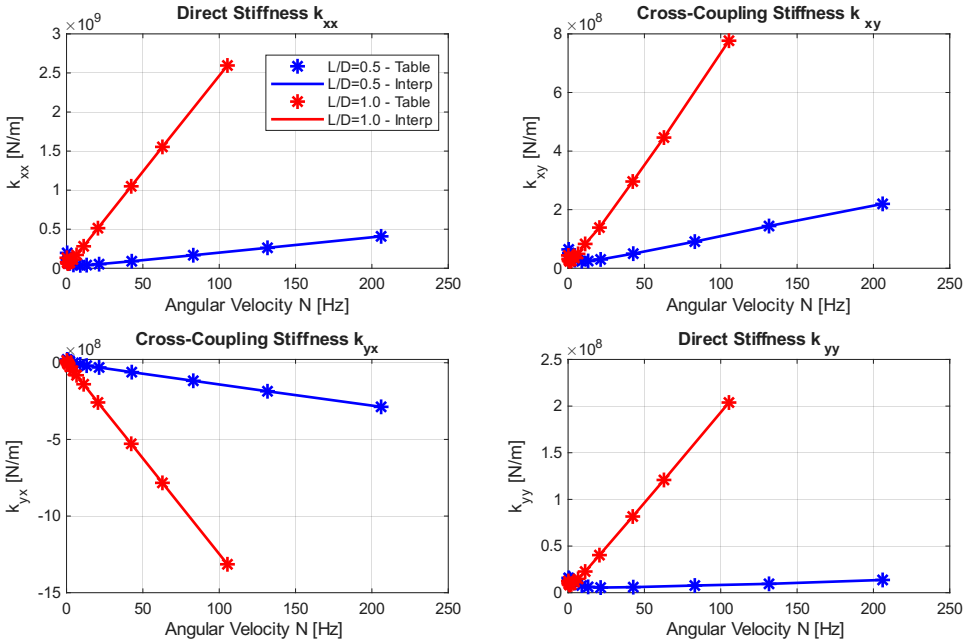


Figure 12: Stiffness Kxx, Kyy, Kxy, Kyx of the hydrodynamic bearings specificied in Table 3 and 4

For the damping coefficients, the $L/D = 1.0$ bearing provides considerably higher direct damping

in both d_{xx} and d_{yy}, which is beneficial for suppressing vibrations near the critical speed. However, the cross-coupling damping terms d_{xy} and d_{yx} are larger in magnitude for the $L/D = 0.5$ bearing. In general, all damping coefficients decrease with increasing angular velocity, which means the system becomes less damped at higher speeds.

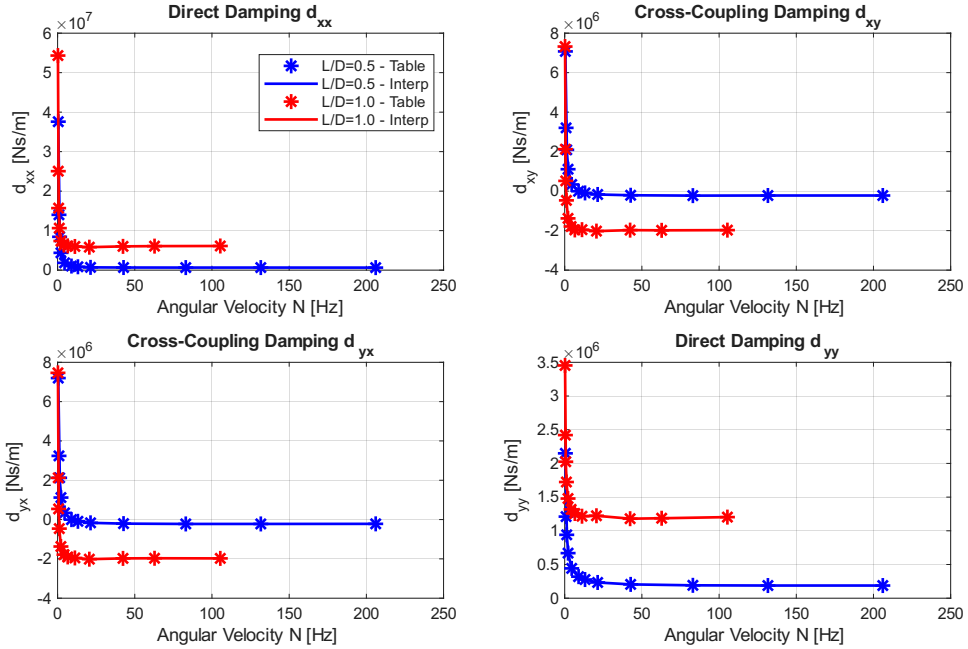


Figure 13: Damping Dxx, Dyy, Dxy, Dyx of the hydrodynamic bearings specificied in Table 3 and 4

Overall, the $L/D = 1.0$ bearing provides significantly higher stiffness and direct damping, making it the more suitable choice for this application and will be used in the following report.

5.2 Prediction of Critical Speeds and Rotor Bearing Stability Limit

The critical speeds of the rotor-bearing system are defined as the rotor speeds at which a natural frequency coincides with the 1X synchronous excitation frequency. Since unbalance forces co-rotate with the shaft, they excite the system at exactly 1X and preferentially drive the forward whirl modes. The critical speeds are therefore identified from the intersections of the forward whirl branches with the 1X synchronous line in the Campbell diagram, and are confirmed by the most pronounced resonance peaks in the unbalance response plot in Figure 16, which occur at the same rotor speeds

From the Campbell diagram in Figure 14, the two critical speeds of System II with the hydrodynamic bearing are $N_{c1} \approx 196\,\text{Hz}$ and $N_{c2} \approx 552\,\text{Hz}$. The operational range between the two critical speeds, in which vibration levels are typically low, is therefore:

$$\Delta N = N_{c2} - N_{c1} \approx 552 - 196 = 356\,\text{Hz} \tag{25}$$

This represents a relatively wide operational range, so the machine can operate well above the first critical speed before approaching the second resonance.

Stability Limit From the stability map in Figure 15, the real parts of the eigenvalues become positive at approximately $N_{\text{thresh}} \approx 198$ Hz, which marks the stability threshold. Beyond this speed, the oil film forces introduce enough energy into the system to cause self-excited vibrations, also described as oil whirl. The system must therefore not exceed this speed in operation.

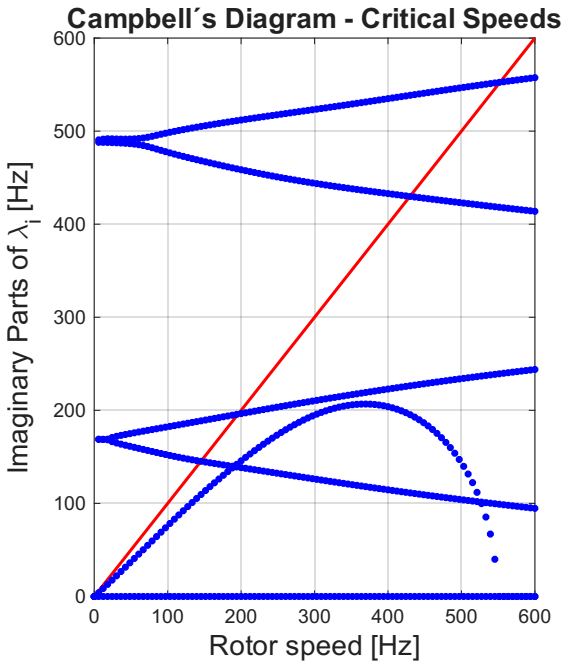


Figure 14: Campbell Diagramm of System II

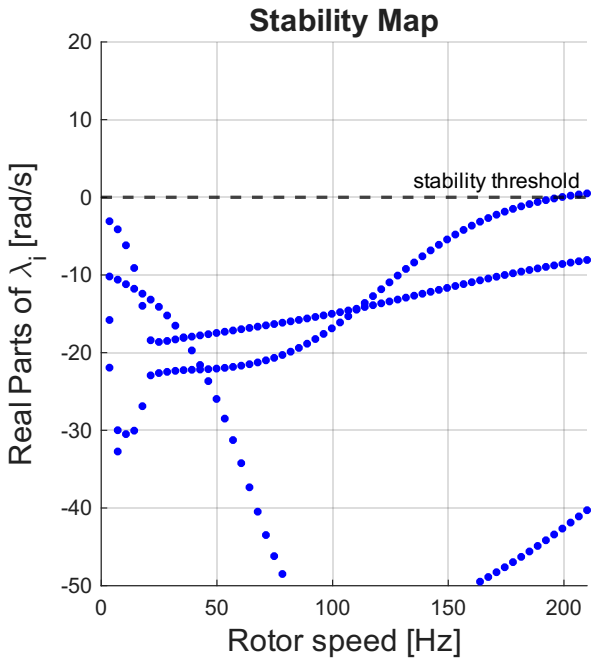


Figure 15: Stability Map of System II

Although the machine can operate stably below the first critical speed, the stability threshold at 198 Hz falls well below the second critical speed of 552 Hz, meaning the oil film instability cuts off the intended operating range between N_{c1} and N_{c2} almost immediately after N_{c1} is passed.

At the stability threshold, the frequency of the unstable vibration is $f_{\text{unstable}} = 138$ Hz. The ratio between the unstable vibration frequency and the rotor angular velocity at the threshold is:

$$\frac{f_{\text{unstable}}}{N_{\text{thresh}}} = \frac{138}{198} \approx 0.7 \tag{26}$$

This ratio of approximately 0.7 is characteristic of oil whirl. For a plain circular bearing, the whirl frequency is typically around 0.5 times the rotor speed. The two-lobe bearing with preload $m_p = 2/3$ and $L/D = 1.0$ shifts this ratio upward, as the geometric preload stiffens the oil film and pushes the instability frequency closer to the rotor speed. The sub-synchronous nature of the instability confirms that the vibration is driven by the oil film rather than unbalance forces, which would produce a synchronous 1X response.

5.3 Unbalance Response

Figure 16 shows the unbalance response at the disc location for three different unbalance levels: 10 g mm, 29.1 g mm, and 35 g mm. Unlike the undamped ball bearing case, the hydrodynamic bearing introduces sufficient damping to limit the resonance amplitude to a finite value.

The maximum permissible unbalance $U_{\max}$ is determined by linear scaling of a test unbalance response. Given a test unbalance U_{test} with peak amplitude $\hat{x}_{\text{test}}$, the scaling factor to meet the 30 µm limit is:

$$U_{\max} = U_{\text{test}} \cdot \frac{x_{\text{limit}}}{\hat{x}_{\text{test}}} \quad \text{with} \quad x_{\text{limit}} = 30\,\mu\text{m} \quad (27)$$

Applying this to the computed response yields a maximum permissible unbalance of $U_{\max} = 29.1$ g mm. As visible in Figure 16, any unbalance beyond this value causes the amplitude to exceed the design limit near the first critical speed, risking rubbing between rotating and stationary parts. It is worth noting that the design specification can now actually be met, in contrast to the purely undamped ball bearing configuration in Section 4.2, where the amplitude was unbounded at resonance.

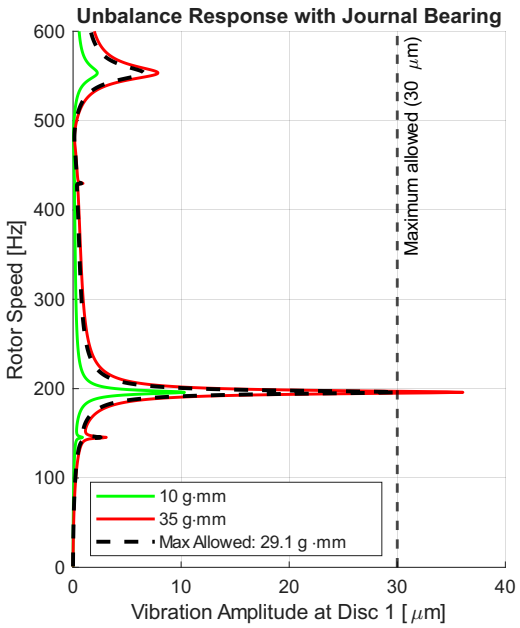


Figure 16: Unbalance of System II with hydrodynamic Bearings

5.4 Engineering Design - Application

Figure 17 shows the stability map for four different bearing combinations when both bearings are hydrodynamic, and Figure 18 presents the corresponding unbalance response. The stability thresholds are summarized in Table 9.

Replacing the ball bearing with a second hydrodynamic bearing introduces an additional source of speed-dependent damping and stiffness into the system, but also an additional source of destabilizing cross-coupling forces that are inherent to hydrodynamic bearings. Whether this substitution improves or deteriorates the stability margin therefore depends critically on the chosen bearing geometry, specifically the L/D ratio at each position.

As shown in Figure 17 and summarized in Table 9, the stability threshold varies significantly across the four combinations. The mixed combination of B1 ($L/D = 1.0$) and B2 ($L/D = 0.5$) achieves the highest stability threshold at approximately 230 Hz, which represents a clear improvement over the original configuration of one ball bearing and one hydrodynamic bearing, which became unstable at approximately 198 Hz. This improvement can be attributed to the asymmetric bearing arrangement: the longer bearing at B1 provides higher load capacity and damping, while the shorter bearing at B2 limits the magnitude of destabilizing cross-coupling forces at the shaft extremity.

In contrast, all remaining combinations perform worse than the original setup. Two identical $L/D = 1.0$ bearings reach a stability threshold of only 170 Hz, two $L/D = 0.5$ bearings only 150 Hz, and the reversed mixed combination of B1 ($L/D = 0.5$) and B2 ($L/D = 1.0$) yields approximately 152 Hz. The degraded performance of symmetric configurations suggests that matching the bearing geometry to the local loading conditions at each shaft position is essential identical bearings on both ends introduce comparable levels of cross-coupling at both supports simultaneously, which destabilizes the system at a lower speed.

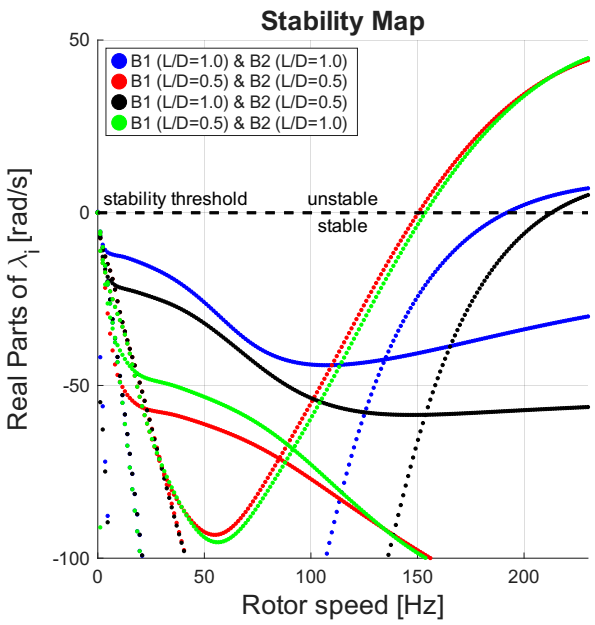


Figure 17: Stability map for System II with different combinations of hydrodynamic bearings at positions B1 and B2.

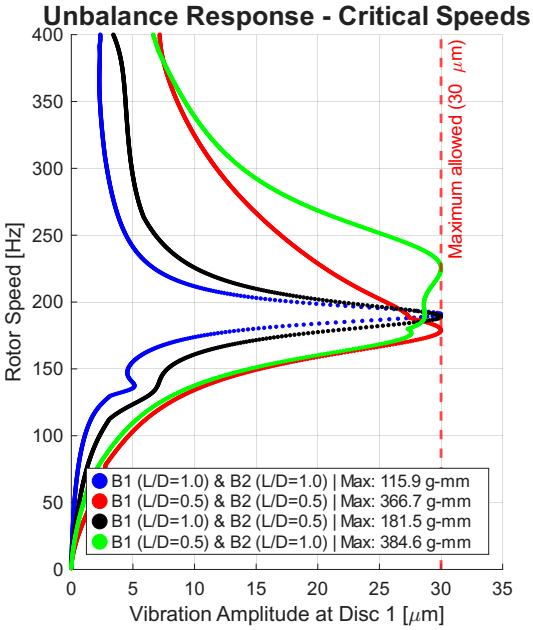


Figure 18: Unbalance for System II with different combinations of hydrodynamic bearings at positions B1 and B2.

Bearing Combination	Stability Threshold [Hz]
B1 (L/D = 1.0) & B2 (L/D = 1.0)	≈ 170 Hz
B1 (L/D = 0.5) & B2 (L/D = 0.5)	≈ 150 Hz
B1 (L/D = 1.0) & B2 (L/D = 0.5)	≈ 230 Hz
B1 (L/D = 0.5) & B2 (L/D = 1.0)	≈ 152 Hz

Table 9: Stability thresholds for different hydrodynamic bearing combinations in System II.

Beyond stability, the unbalance response provides additional insight for the final bearing selection. Interestingly, a clear trade-off emerges: the combinations with the highest stability thresholds are also the least tolerant to manufacturing unbalance. The combination of B1 (L/D = 1.0) and B2 (L/D = 0.5), which achieves the best stability margin at 230 Hz, allows a maximum unbalance of only 181.5 g·mm before exceeding the 30 µm vibration limit. In contrast, the combination of B1 (L/D = 0.5) and B2 (L/D = 1.0) tolerates up to 384.6 g·mm of unbalance, but at the cost of a significantly lower stability threshold of 152 Hz. Therefore, the final bearing selection depends on the manufacturing precision of the rotor. If tight tolerances can be guaranteed, the B1 (L/D = 1.0) and B2 (L/D = 0.5) combination is the superior choice, otherwise, a compromise towards higher unbalance tolerance must be accepted at the expense of the stability margin.

In conclusion, replacing the ball bearing with a second hydrodynamic bearing does not universally improve the stability margin. Only the specific combination of B1 (L/D = 1.0) and B2 (L/D = 0.5) yields an improvement over the original configuration, while all other combinations reduce the stability threshold below 198 Hz. This highlights that the design of a double hydrodynamic bearing system requires careful matching of bearing proportions to shaft position, and that symmetric configurations should be avoided for this particular rotor geometry.

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6 References and Declaration of AI-usage

6.1 Declaration of AI Usage

This report was written with the assistance of Artificial Intelligence. Specifically, AI was utilized for [e.g., structuring the LaTeX code, refining technical descriptions in Section 2, and grammar correction]. The final content was reviewed and verified by the author.

6.2 References and Methodology

The mathematical derivations for the flexible shaft and disc dynamics are based on the course materials from *Dynamics of Machinery*, as taught by Professor Ilmar Ferreira Santos.

6.2.1 For the static load on the bearing and the sentence of Steiner:

N. Rajapakse, D. Gross, W. Hauger, J. Schröder, and W. A. Wall, *Engineering Mechanics 1: Statics*, 1st ed. Berlin, Heidelberg: Springer, 2009. doi: 10.1007/978-3-540-89937-2.

D. Gross, W. Hauger, J. Schröder, W. A. Wall, and J. Bonet, *Engineering Mechanics 2: Mechanics of Materials*, 2nd ed. Berlin, Heidelberg: Springer, 2018. doi: 10.1007/978-3-662-56272-7.

7 Appendix

7.1 Experimental Data

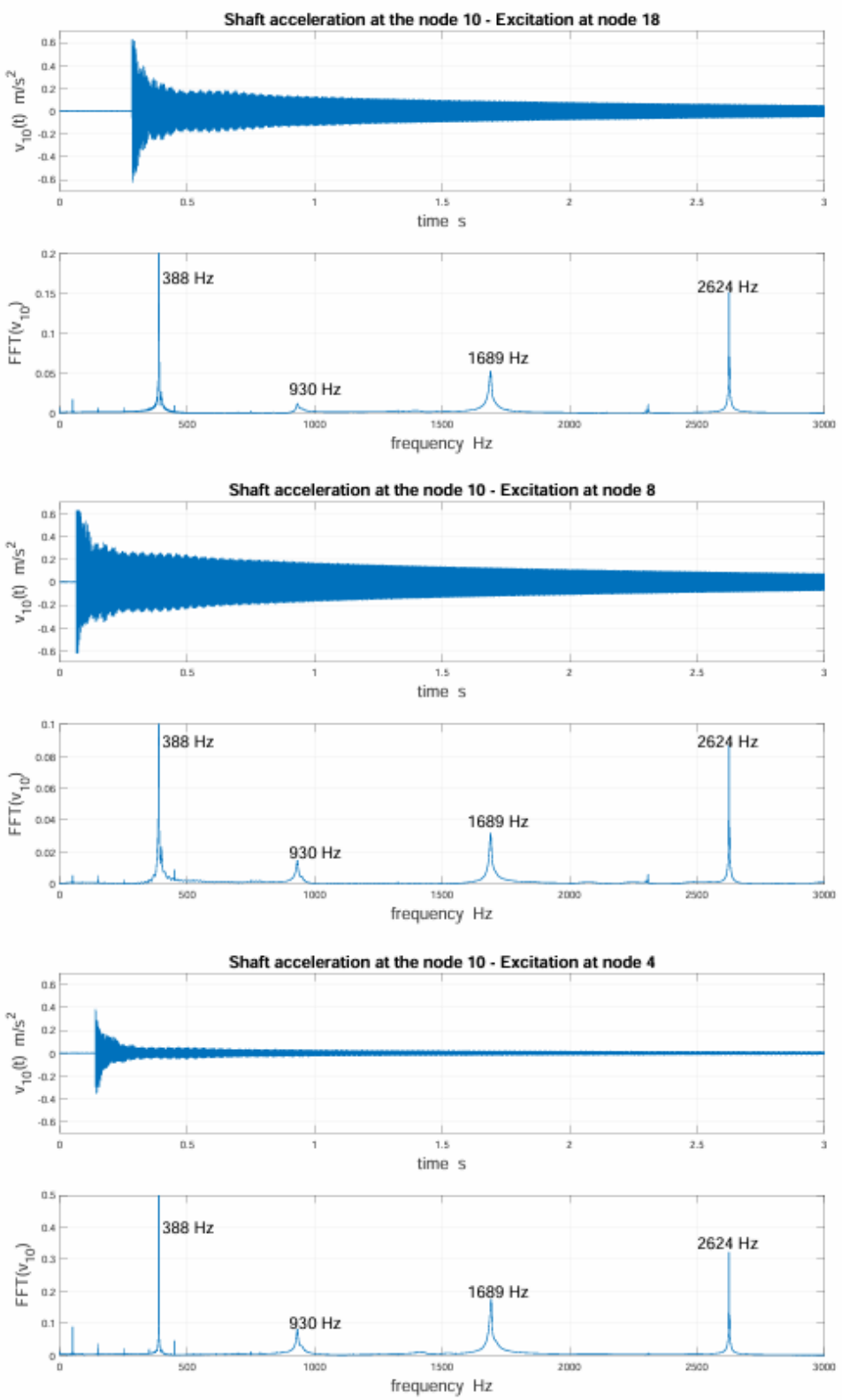


Figure 19: Partial results of experimental modal testing–flexible shaft in a"free-free"condition and its natural frequencies. Three time series and their Fast Fourier Transformation (FFT) obtained when the shaft is excited by the impact hammer at nodes 18, 8, and 4 and its lateral acceleration response is measured using the accelerometer placed at the node 10.

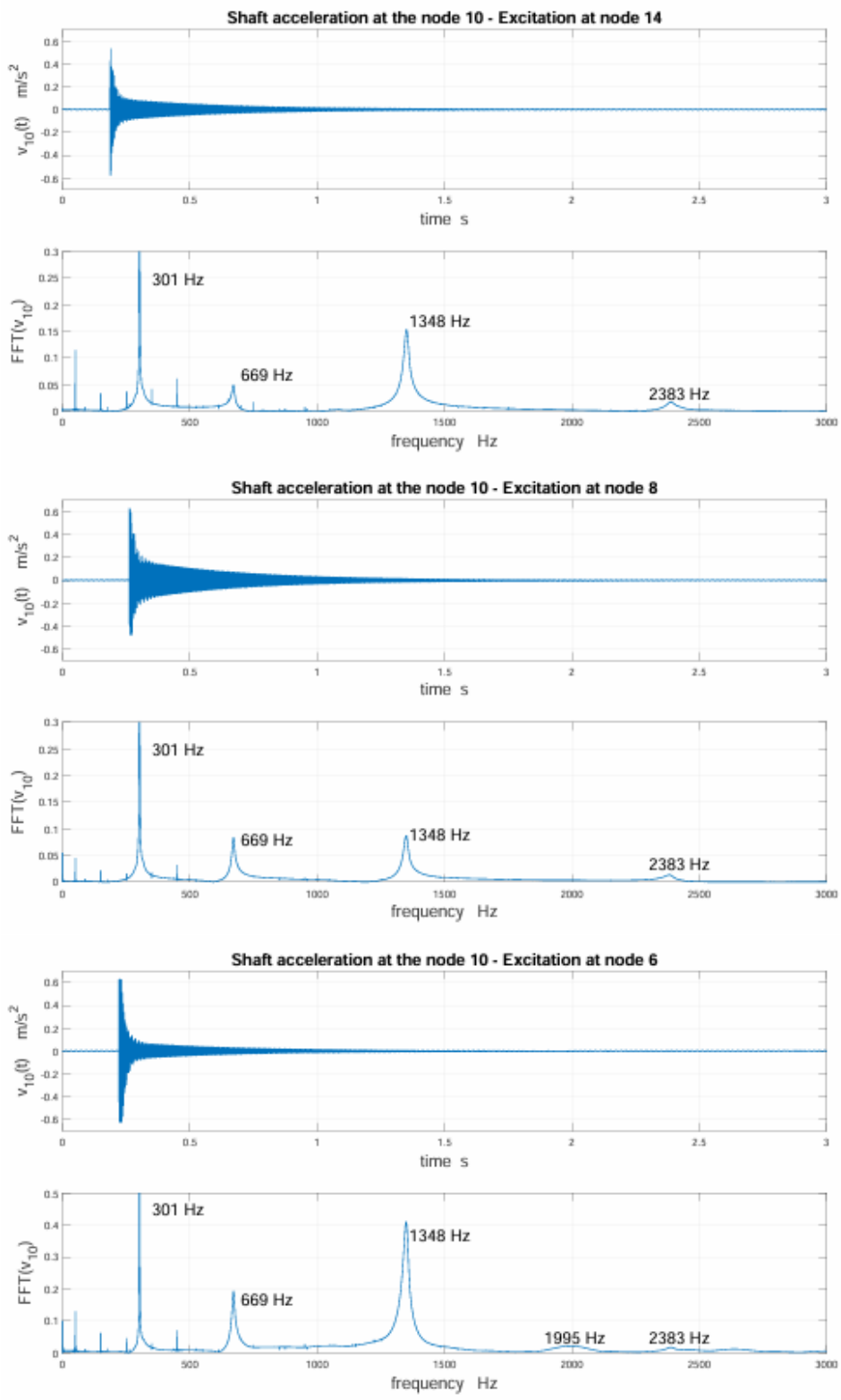


Figure 20: Partial results from experimental modal testing–flexible shaft coupled to the rigid disc of 100mm thickness in a ”free-free” condition and its natural frequencies. Three time series and their Fast Fourier Transformation (FFT) obtained when the shaft is excited by the impact hammer at nodes 14, 8, and 6 and its lateral acceleration response is measured using the accelerometer placed at the node 10.

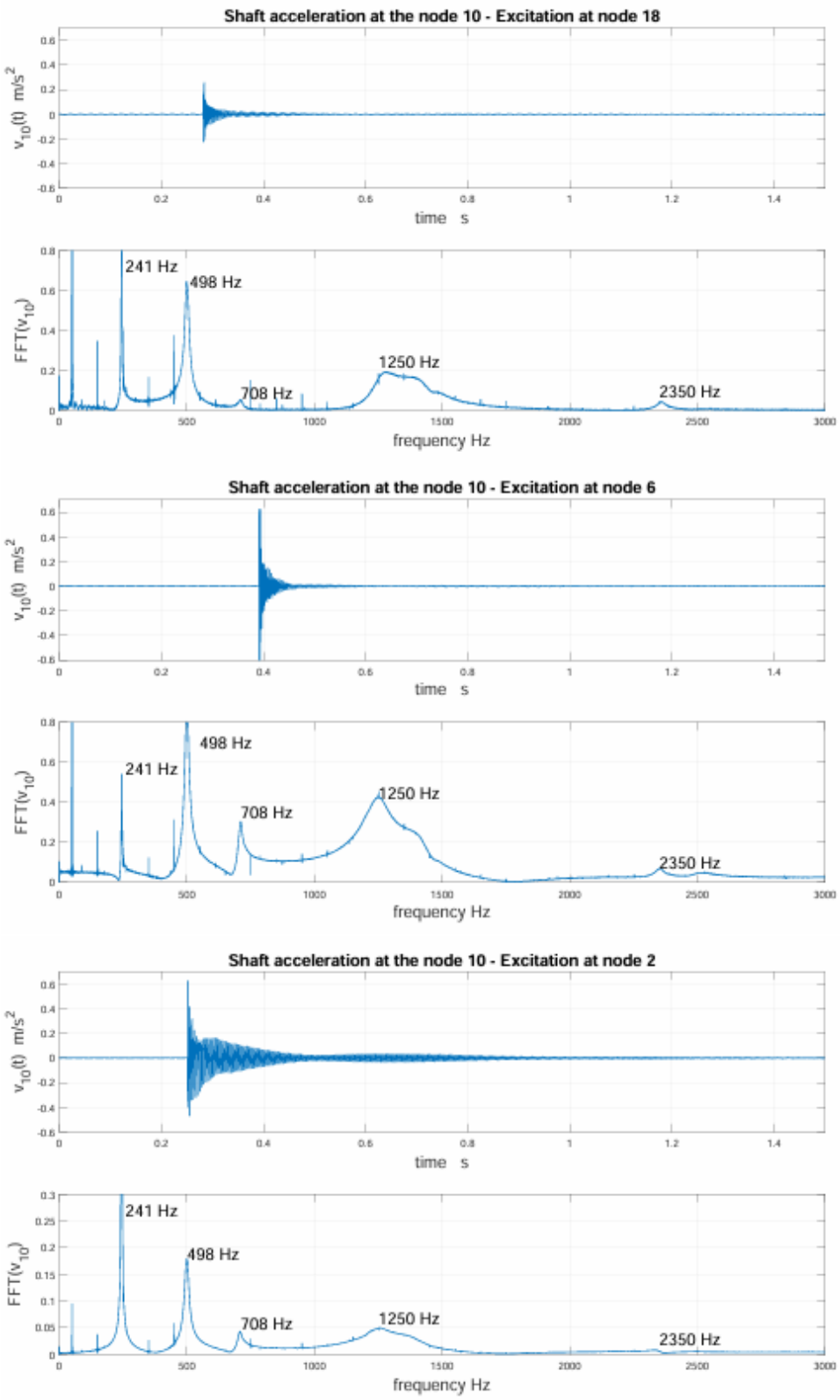


Figure 21: Partial results from experimental modal testing–flexible shaft coupled to the two rigid discs of 100 mm and 80 mm tihckness in a ”free-free” condition and its natural frequencies. Three time series and their Fast Fourier Transformation (FFT) obtained when the shaft is excited by the impact hammer at nodes 18, 6, and 2 and its lateral acceleration response is measured using the accelerometer placed at the node 10.

7.2 Used Hydrodynamic Bearings

```
% \begin{minted}[breaklines, fontsize=\footnotesize]{matlab}
% Bearing Properties (PAGE 41 of JOURNAL BEARING DATA BOOK)
% Two-lobe bearing, L/D = 0.5, mp = 1/2
%
%        S       E    Phi    Q     P     T       Kxx  Kxy   Kyx    Kyy  Bxx     Bxy     Byx     Byy
%
Table3=[4.79   0.0556 87.2 0.292 0.530 104     61.5 33.1  -43.3   2.06  123    -43.3   -42.5    36.6
       3.06   0.0868 87.2 0.293 0.524  65.9   39.4 21.7  -28.0   1.43   80.0  -27.7   -27.7    23.4
       1.93   0.137  86.9 0.296 0.520  41.6   25.1 13.7  -17.7   1.16   50.2  -17.9   -17.5    15.0
       0.993  0.262  85.9 0.312 0.502  21.4   13.5  7.50  -9.20  0.897  26.5   -8.47   -8.34    8.28
       0.496  0.438  84.4 0.343 0.469  10.8    7.88 4.52  -4.60  0.831  14.0   -3.42   -3.36    4.71
       0.306  0.608  80.8 0.370 0.441   6.99   6.06 3.73  -2.82  0.940  10.5   -1.13   -1.11    3.45
       0.209  0.718  77.2 0.402 0.404   4.91   5.66 3.60  -1.67  1.16    8.74  -0.0503 -0.0240  2.72
       0.105  0.888  68.3 0.433 0.364   2.73   6.51 4.19  -0.115 1.57    7.86   1.44    1.45    1.89
       0.0507 0.970  56.8 0.436 0.353   1.61   9.47 5.46   0.945 1.94    9.03   2.29    2.31    1.38
       0.0303 0.995  48.6 0.424 0.359   1.12  13.2  6.45   1.72  2.12    10.4   2.58    2.62    1.16
       0.0207 1.01   42.5 0.407 0.370   0.891 17.2  7.81   2.33  2.24    11.8   2.70    2.73    1.02
       0.0100 1.01   34.1 0.372 0.400   0.590 29.6  9.68   3.28  2.39    15.3   2.88    2.93    0.875];

% Bearing Properties (PAGE 53 of JOURNAL BEARING DATA BOOK)
% Two-lobe bearing, L/D = 1.0, mp = 2/3
%
%        S       E    Phi    Q     P     T    Kxx  Kxy   Kyx    Kyy  Bxx   Bxy    Byx   Byy
%
Table4=[4.90   0.0172 81.2 0.218 0.379 133   391  117  -198   30.7  609  -197   -197   120
        2.92   0.0289 81.2 0.218 0.379 78.8  234  67.2 -118   18.2  361  -118   -117   70.5
        1.97   0.0430 81.2 0.218 0.379 53.2  158  44.6 -79.6  12.3  241  -79.2  -79.2  47.3
        0.956  0.0898 81.2 0.219 0.378 25.5  77.3 20.9 -39.0  6.07  113  -39.4  -39.2  23.8
        0.521  0.165  81.0 0.219 0.377 14.1  42.3 12.5 -21.1  3.42  63.5 -20.7  -20.7  12.9
        0.305  0.277  80.7 0.222 0.372 8.26  25.6 7.40 -12.3  2.23  38.1 -12.2  -11.8  7.80
        0.192  0.427  80.0 0.228 0.363 5.33  16.8 5.30 -7.80  1.71  24.7 -7.00  -6.98  5.15
        0.100  0.655  78.3 0.239 0.344 2.93  10.7 3.82 -3.97  1.34  15.0 -2.81  -2.79  3.01
        0.0513 0.925  74.0 0.256 0.311 1.66  8.92 3.55 -1.61  1.24  11.1 -0.490 -0.481 1.80
        0.0313 1.07   69.1 0.263 0.292 1.15  9.61 4.02 -0.401 1.36  10.0 0.330  0.345  1.29
        0.0202 1.15   64.0 0.265 0.281 0.865 11.7 4.80 0.570  1.56  10.3 0.873  0.873  0.995
        0.0104 1.15   53.9 0.247 0.296 0.559 20.2 6.60 2.24   1.90  11.5 1.55   1.58   0.731];
```

7.3 Technical Documentation of the Investigated Components

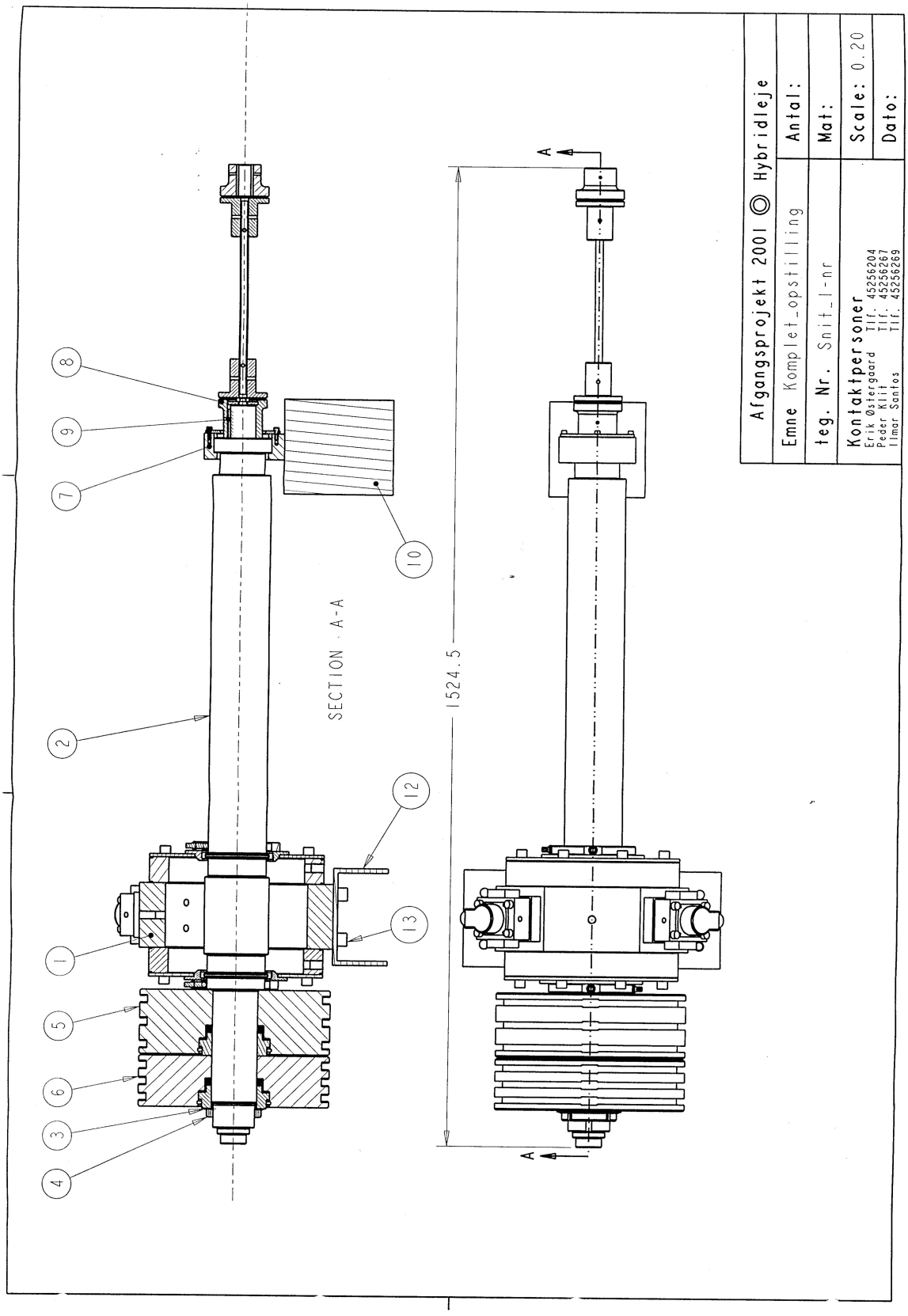


Figure 22: Setup of the Testunit

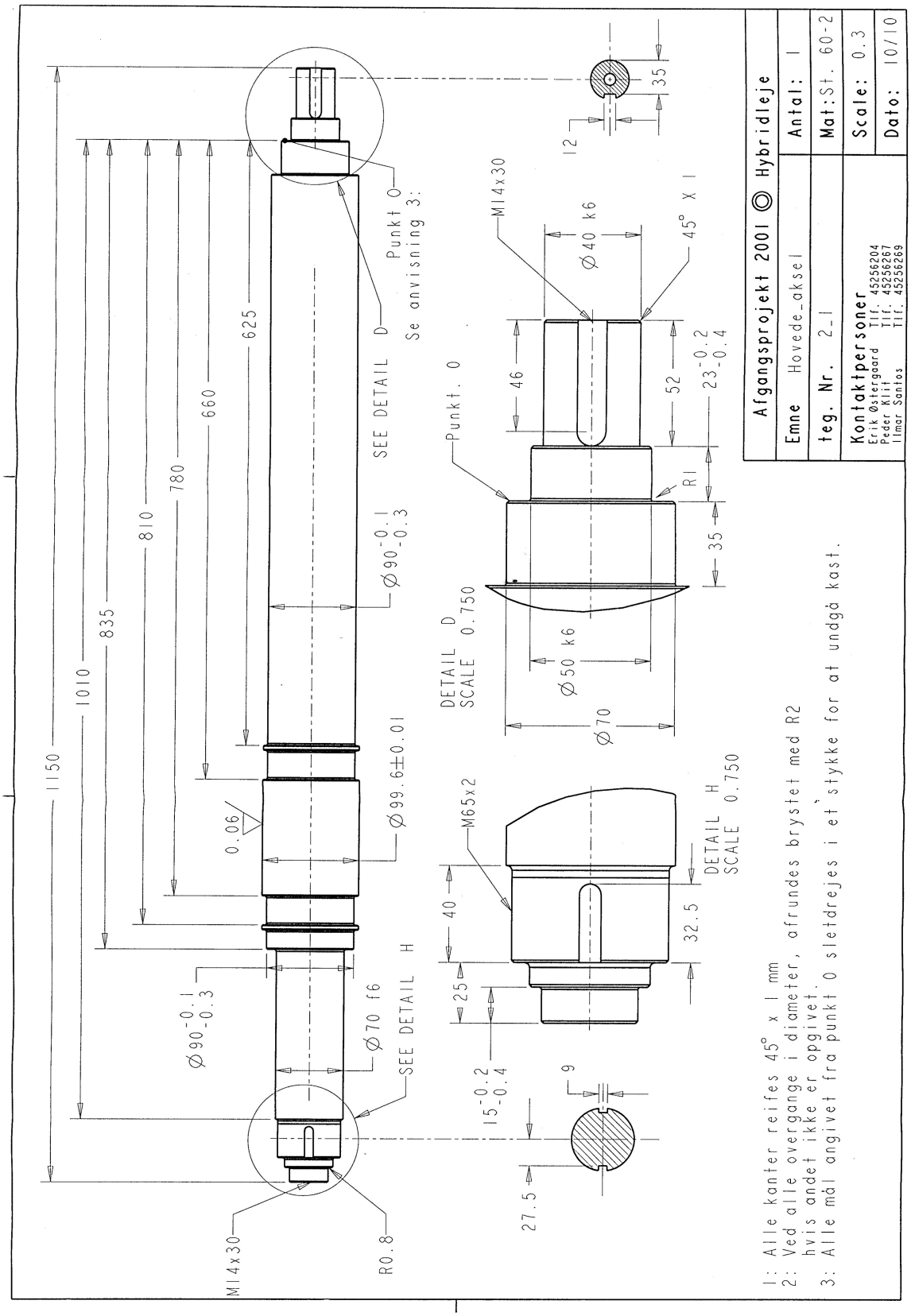


Figure 23: Shaft Dimensions

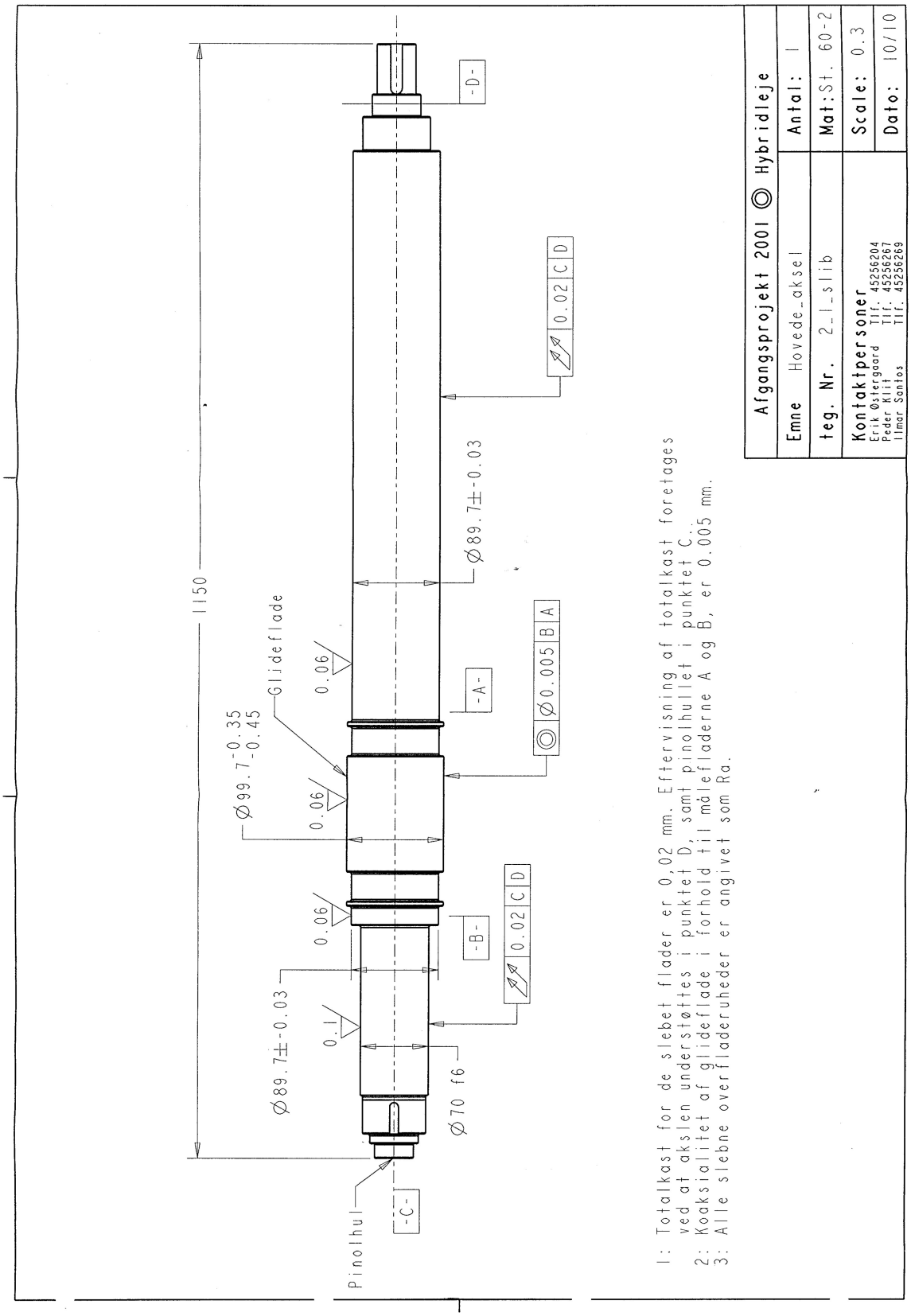


Figure 24: Shaft Dimensions

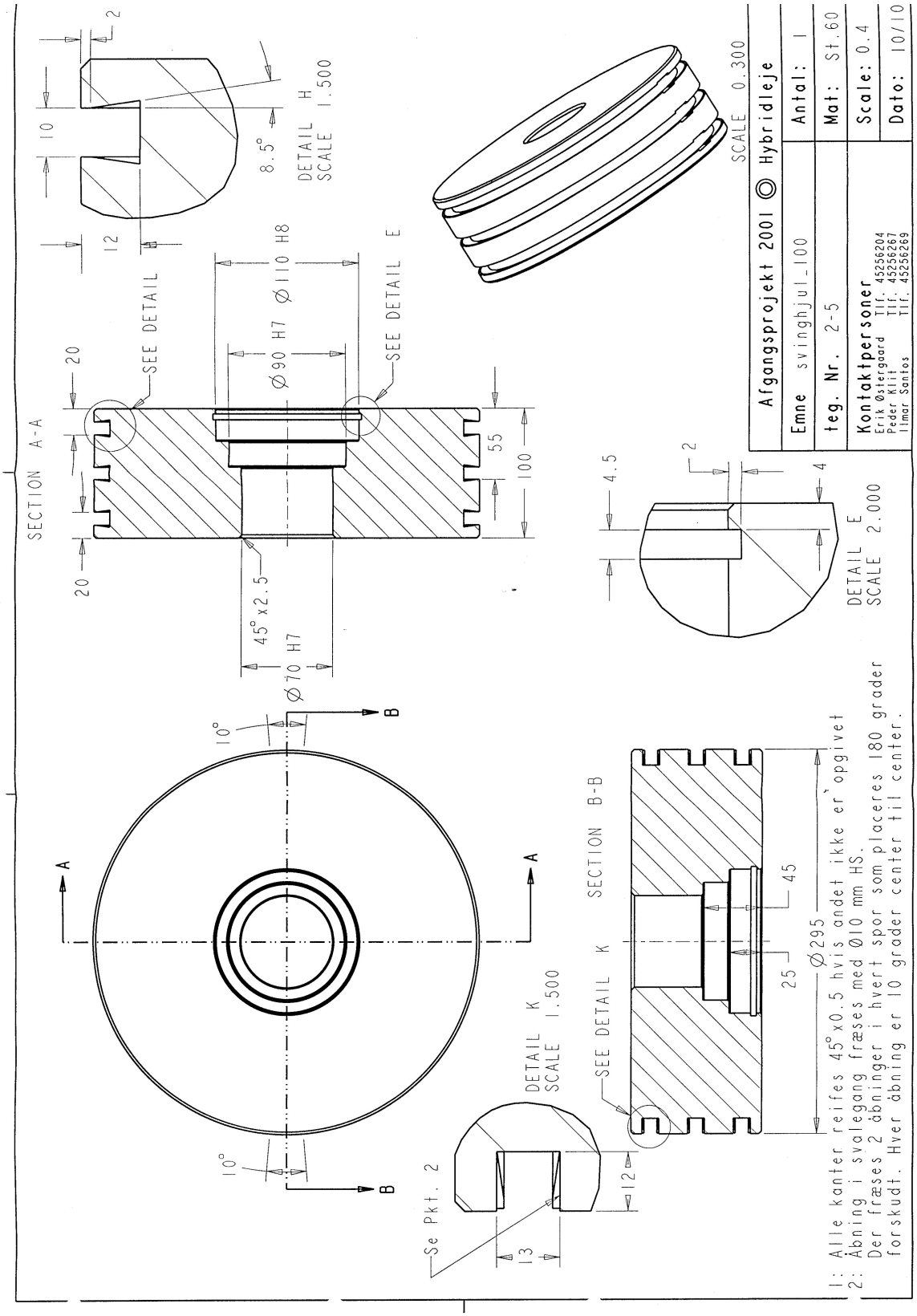


Figure 25: Dimensions of Disc 1

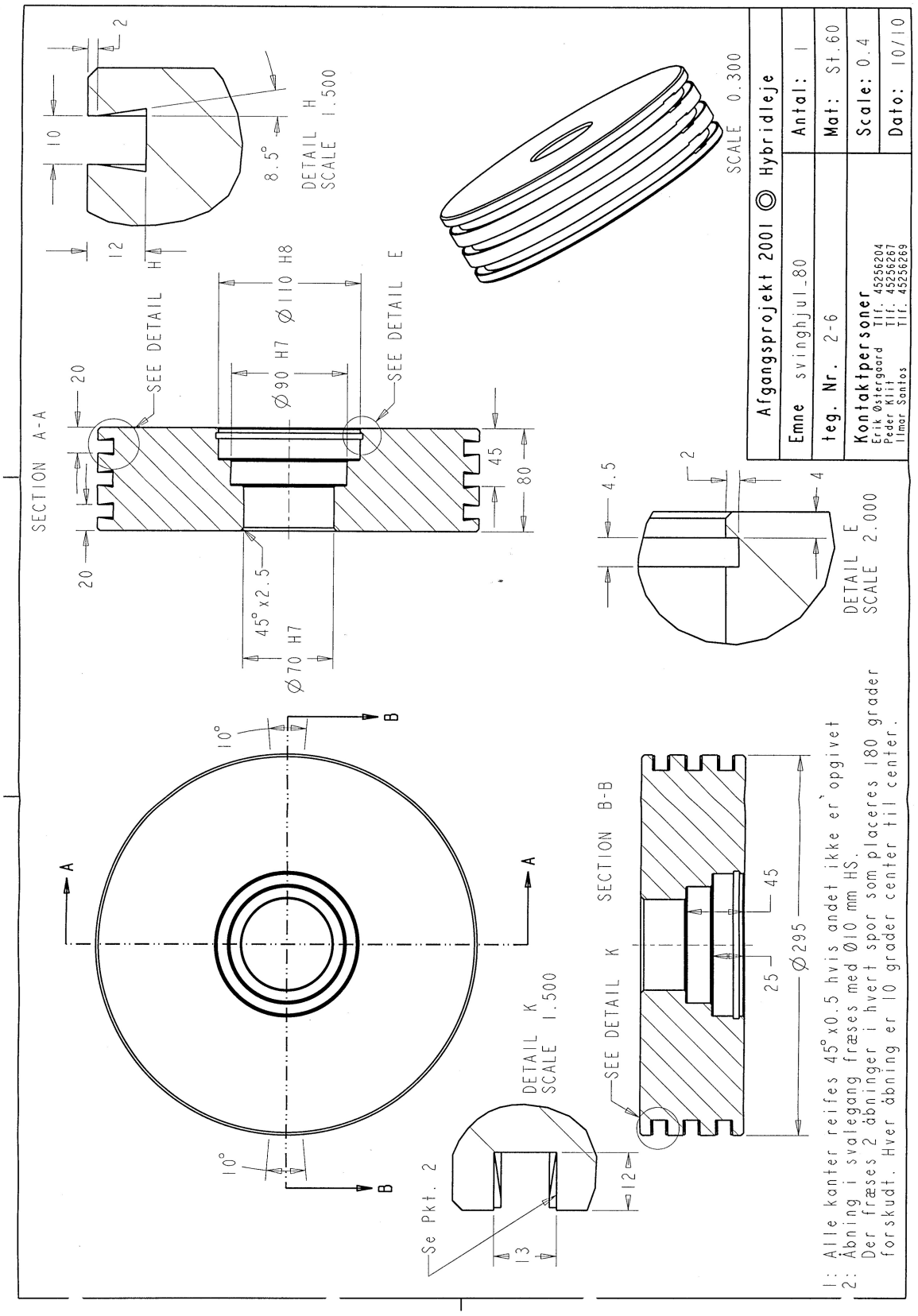


Figure 26: Dimensions of Disc 2